

Geometry III

Key definitions

1 Curvature, Gauss map and fundamental forms

In this section we collect the basic definitions and results about curvature of plane curves, the Gauss map and the first and second fundamental forms of a surface in $\mathbb{R}^3$. We also introduce ruled, developable and (strictly) convex surfaces and state the fundamental theorem of surface theory.

1.1 Curvature of plane curves

Let $\gamma : (a, b) \rightarrow \mathbb{R}^2$ be a smooth regular plane curve.

Definition 1 (Arc-length parametrisation). *We say that γ is parametrised by arc-length if*

$$\|\gamma'(s)\| = 1 \quad \text{for all } s \in (a, b).$$

In that case $\gamma'(s)$ is a unit tangent vector along the curve.

For a unit speed curve, differentiating $\langle \gamma'(s), \gamma'(s) \rangle = 1$ gives $\langle \gamma'(s), \gamma''(s) \rangle = 0$, so $\gamma''(s)$ is always orthogonal to $\gamma'(s)$ and therefore normal to the curve.

Definition 2 (Curvature of a plane curve). *Let $\gamma : (a, b) \rightarrow \mathbb{R}^2$ be a plane curve parametrised by arc-length. A unit normal field $n_\gamma(s)$ along γ is a smooth map $(a, b) \rightarrow \mathbb{R}^2$ such that*

$$\|n_\gamma(s)\| = 1, \quad \langle \gamma'(s), n_\gamma(s) \rangle = 0$$

for all s . We choose $n_\gamma(s)$ so that $(\gamma'(s), n_\gamma(s))$ is a positively oriented basis of $\mathbb{R}^2$ (the “turn left” normal).

The curvature of γ is the function $\kappa : (a, b) \rightarrow \mathbb{R}$ such that

$$\gamma''(s) = \kappa(s) n_\gamma(s).$$

The quantity $\rho(s) := 1/|\kappa(s)|$ is called the radius of curvature.

Equivalently, if we write the unit tangent as

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s)),$$

where $\theta(s)$ is the angle between $\gamma'(s)$ and the x -axis, then

$$\gamma''(s) = \theta'(s) (-\sin \theta(s), \cos \theta(s)) = \theta'(s) n_\gamma(s),$$

so

$$\kappa(s) = \theta'(s),$$

i.e. curvature is the rate of change of the tangent angle with respect to arc-length.

Example 1.1.

1. Straight line: $\gamma(s) = (s, 0)$ is unit speed, with $\gamma''(s) = 0$, so $\kappa(s) \equiv 0$.
2. Circle of radius R , traversed counterclockwise:

$$\gamma(s) = (R \cos(s/R), R \sin(s/R)).$$

Then $\gamma'(s)$ has unit length and $\gamma''(s) = \frac{1}{R}n_\gamma(s)$, so $\kappa(s) \equiv 1/R$.

3. Parabola $y = x^2$, with parametrisation $\tilde{\gamma}(t) = (t, t^2)$. Using the general formula

$$\kappa(t) = \frac{x'(t)y''(t) - y'(t)x''(t)}{((x'(t))^2 + (y'(t))^2)^{3/2}},$$

we obtain

$$\kappa(t) = \frac{2}{(1 + 4t^2)^{3/2}} > 0,$$

largest at $t = 0$ and decaying as $|t| \rightarrow \infty$.

1.2 Surfaces, the Gauss map and the second fundamental form

Let $\Sigma \subset \mathbb{R}^3$ be an oriented smooth surface, with unit normal field $n : \Sigma \rightarrow S^2$.

Definition 3 (Gauss map). *The Gauss map of Σ is the map*

$$N : \Sigma \rightarrow S^2, \quad N(p) = n(p),$$

assigning to each point its unit normal vector (chosen consistently with the orientation).

Definition 4 (Shape operator and second fundamental form). *The shape operator at $p \in \Sigma$ is the linear map $S_p : T_p\Sigma \rightarrow T_p\Sigma$ defined by*

$$S_p(w) = -(dN)_p(w).$$

The second fundamental form at p is the symmetric bilinear form

$$\mathbb{I}_p(u, w) = \langle S_p(u), w \rangle = -\langle (dN)_p(u), w \rangle, \quad u, w \in T_p\Sigma.$$

In local coordinates $\sigma(u, v)$, with first fundamental form coefficients

$$E = \langle \sigma_u, \sigma_u \rangle, \quad F = \langle \sigma_u, \sigma_v \rangle, \quad G = \langle \sigma_v, \sigma_v \rangle,$$

and second fundamental form coefficients

$$L = \langle \sigma_{uu}, n \rangle, \quad M = \langle \sigma_{uv}, n \rangle, \quad N = \langle \sigma_{vv}, n \rangle,$$

one has

$$K = \det S_p = \frac{LN - M^2}{EG - F^2},$$

the Gauss curvature at p .

Definition 5 (Normal curvature). *Let $p \in \Sigma$ and let $v \in T_p \Sigma$ be a unit tangent vector. Consider the plane $P = p + \text{span}(v, n(p))$ and the plane curve γ_v given by the intersection of Σ with P , parametrised by arc-length and passing through p with $\gamma'_v(0) = v$. The normal curvature of Σ at p in the direction v is the curvature $\kappa_v(0)$ of the plane curve γ_v at p .*

Proposition 1.2 (Normal curvature and second fundamental form). *With notation as above, the normal curvature in the direction v is*

$$\kappa_v(0) = \mathbb{I}_p(v, v).$$

Idea of proof. Along γ_v we have $\langle \gamma'_v(s), n(\gamma_v(s)) \rangle = 0$. Differentiating at $s = 0$ gives

$$\langle \gamma''_v(0), n(p) \rangle + \langle \gamma'_v(0), (dn)_p(\gamma'_v(0)) \rangle = 0.$$

Using $\gamma'_v(0) = v$ and the definition of $\mathbb{I}_p$ we obtain

$$\langle \gamma''_v(0), n(p) \rangle = \mathbb{I}_p(v, v).$$

As a plane curve in the oriented plane P , γ_v has curvature $\kappa_v(0)$ determined by $\gamma''_v(0) = \kappa_v(0)n(p)$. Taking the inner product with $n(p)$ yields $\kappa_v(0) = \langle \gamma''_v(0), n(p) \rangle = \mathbb{I}_p(v, v)$. $\square$

1.3 Ruled, developable and convex surfaces

Definition 6 (Ruled surface). *A smooth surface $\Sigma \subset \mathbb{R}^3$ is called ruled if through every point $p \in \Sigma$ there passes at least one straight line (a ruling) contained entirely in Σ .*

A typical local parametrisation is

$$\sigma(u, v) = c(u) + v d(u),$$

where $c(u)$ is a space curve (the directrix) and $d(u)$ is a direction vector. For fixed u , the map $v \mapsto \sigma(u, v)$ is a straight line.

Example 1.3.

1. Plane: $\sigma(u, v) = (u, v, 0)$.
2. Cylinder: $\sigma(u, v) = (\cos u, \sin u, v)$.
3. Cone: $\sigma(u, v) = (v \cos u, v \sin u, v)$.
4. Helicoid: $\sigma(u, v) = (v \cos u, v \sin u, au)$.
5. Hyperbolic paraboloid: $\sigma(u, v) = (u, v, uv)$.

Definition 7 (Developable surface). *A regular surface $\Sigma \subset \mathbb{R}^3$ is called developable if it is locally isometric to the plane (it can be unfolded onto $\mathbb{R}^2$ without stretching). For regular surfaces this is equivalent to vanishing Gauss curvature:*

$$\Sigma \text{ developable} \iff K \equiv 0.$$

Among ruled surfaces, a smooth ruled surface with $K = 0$ is locally a piece of a plane, cylinder, cone or a tangent developable of a space curve. Ruled surfaces with $K \neq 0$ (such as the helicoid or the hyperbolic paraboloid, which have $K < 0$) are *non-developable*.

Definition 8 (Convex and strictly convex surfaces). A set $R \subset \mathbb{R}^3$ is convex if for any $x, y \in R$ the segment $[x, y]$ lies in R . A compact smooth surface $\Sigma \subset \mathbb{R}^3$ is called convex if it is the boundary of a convex region R .

We say $\Sigma = \partial R$ is strictly convex if R is convex and for any $x, y \in R$ one has

$$[x, y] \subset R, \quad [x, y] \cap \Sigma \subset \{x, y\},$$

i.e. no nontrivial line segment lies on the boundary.

Strict convexity implies convexity, but not conversely (cylinders, cubes and prisms are convex but not strictly convex). For smooth strictly convex surfaces, the Gauss map turns out to be injective and, under the additional assumption $K > 0$, a global diffeomorphism onto S^2 . In that case one obtains the “convex Gauss–Bonnet theorem”

$$\int_{\Sigma} K \, dA = 4\pi.$$

1.4 Matrix formulation and the fundamental theorem

Let $\sigma : V \rightarrow \Sigma \subset \mathbb{R}^3$ be an allowable parametrisation, with $V \subset \mathbb{R}^2$ a connected open set with coordinates (u, v) . Write E, F, G and L, M, N for the first and second fundamental form coefficients as above, and let n be the unit normal along σ .

Define

$$\alpha_{ijk} = \langle \sigma_i, \sigma_{jk} \rangle, \quad i, j, k \in \{u, v\},$$

where subscripts denote partial derivatives. Differentiating E, F, G and using symmetry of the inner product, one finds that all α_{ijk} are determined by E, F, G and their first derivatives.

Let

$$P = (\sigma_u \quad \sigma_v \quad n)$$

be the 3×3 matrix whose columns are σ_u, σ_v and n . Define

$$C = \begin{pmatrix} E & F & 0 \\ F & G & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$D_1 = \begin{pmatrix} \alpha_{uuu} & \alpha_{uvu} & -L \\ \alpha_{vuu} & \alpha_{vvu} & -M \\ L & M & 0 \end{pmatrix}, \quad D_2 = \begin{pmatrix} \alpha_{uvv} & \alpha_{vvv} & -M \\ \alpha_{vvv} & \alpha_{vvv} & -N \\ M & N & 0 \end{pmatrix}.$$

A direct computation using $\langle \sigma_u, n \rangle = \langle \sigma_v, n \rangle = 0$ and $\langle n, n \rangle = 1$ shows

$$P^t P = C.$$

Similarly, $P_u = (\sigma_{uu} \quad \sigma_{uv} \quad n_u)$ gives

$$P^t P_u = D_1,$$

and $P_v = (\sigma_{uv} \quad \sigma_{vv} \quad n_v)$ gives

$$P^t P_v = D_2.$$

Let

$$A = C^{-1} D_1, \quad B = C^{-1} D_2.$$

Using $C = P^t P$ one finds

$$A = P^{-1} P_u, \quad B = P^{-1} P_v,$$

hence

$$P_u = P A, \quad P_v = P B.$$

Taking transposes,

$$P_u^t = A^t P^t, \quad P_v^t = B^t P^t.$$

Thus each column ξ of P^t satisfies the linear first-order PDE system

$$\xi_u = A^t \xi, \quad \xi_v = B^t \xi.$$

Theorem 1.4 (Fundamental theorem of surface theory, uniqueness part). *Suppose two connected smooth surfaces $\Sigma_1, \Sigma_2 \subset \mathbb{R}^3$ have the same first and second fundamental forms on a parameter domain $V \subset \mathbb{R}^2$. Then Σ_1 and Σ_2 are congruent, i.e. there exists a rigid motion of $\mathbb{R}^3$ mapping Σ_1 onto Σ_2 .*

Idea of proof. The fundamental forms determine the matrices C, D_1, D_2 and hence the matrices A, B . For each surface, the associated matrix P (whose columns are σ_u, σ_v, n) satisfies $P_u = P A$, $P_v = P B$, so the columns of P^t satisfy the same linear system $\xi_u = A^t \xi$, $\xi_v = B^t \xi$.

By composing with a rigid motion if necessary, we can arrange that at some point $(u_0, v_0) \in V$ the positions and frames agree: $\sigma^{(1)}(u_0, v_0) = \sigma^{(2)}(u_0, v_0)$ and $P^{(1)}(u_0, v_0) = P^{(2)}(u_0, v_0)$. Assuming uniqueness of solutions for the linear PDE system with such initial data, it follows that $P^{(1)}(u, v) = P^{(2)}(u, v)$ for all $(u, v) \in V$. In particular the first two columns (the partial derivatives of the parametrisations) coincide, so the two surfaces differ only by the initial rigid motion. $\square$

Problem 3: Partial converse to Clairaut's relation

Problem. Let $\Sigma \subset \mathbb{R}^3$ be a smooth surface of revolution. Suppose a smooth curve $\gamma : (a, b) \rightarrow \Sigma$ satisfies

$$\rho(t) \cos \theta(t) = \text{constant},$$

where $\rho(t)$ is the distance from $\gamma(t)$ to the axis of revolution and $\theta(t)$ is the angle between γ and the parallel at $\gamma(t)$. Assume that there is no positive-length interval on which γ coincides with a parallel. Show that γ is a geodesic.

Theory. A surface of revolution is parametrized by

$$X(u, v) = (\rho(u) \cos v, \rho(u) \sin v, z(u)),$$

with induced metric

$$g = E(u) du^2 + G(u) dv^2, \quad E(u) = \rho'(u)^2 + z'(u)^2, \quad G(u) = \rho(u)^2.$$

For a unit-speed curve $(u(s), v(s))$, the tangent has components u', v' and satisfies $E u'^2 + G v'^2 = 1$. The angle θ between γ and the parallel (direction ∂_v) is determined by

$$\cos \theta = g\left(\dot{\gamma}, \frac{1}{\sqrt{G}} \partial_v\right) = \sqrt{G} v' = \rho v'.$$

Hence $\rho \cos \theta = Gv'$.

Solution. Reparametrize γ by arc length s , so that

$$E(u)u'^2 + G(u)v'^2 = 1.$$

The Clairaut condition reads

$$G(u(s)) v'(s) = c, \quad c \in \mathbb{R}. \quad (1)$$

Differentiating,

$$0 = \frac{d}{ds}(Gv') = G_u u' v' + Gv'',$$

so

$$v'' + \frac{G_u}{G} u' v' = 0. \quad (2)$$

For $g = E(u) du^2 + G(u) dv^2$ the Christoffel symbols with index 2 are $\Gamma_{12}^2 = \Gamma_{21}^2 = \frac{1}{2G} G_u$, so the v -geodesic equation

$$v'' + 2\Gamma_{12}^2 u' v' = 0$$

is exactly (2).

Now differentiate the unit-speed condition:

$$0 = \frac{d}{ds}(Eu'^2 + Gv'^2) = E_u u' u'^2 + 2Eu' u'' + G_u u' v'^2 + 2Gv' v''.$$

Using (2) to eliminate v'' , we obtain (for $u' \neq 0$)

$$2Eu'' + E_u u'^2 - G_u v'^2 = 0,$$

or

$$u'' + \frac{E_u}{2E} u'^2 - \frac{G_u}{2E} v'^2 = 0. \quad (3)$$

For this metric, $\Gamma_{11}^1 = \frac{E_u}{2E}$ and $\Gamma_{22}^1 = -\frac{G_u}{2E}$, so (3) is exactly the u -geodesic equation. On any interval where $u' \neq 0$, γ satisfies both geodesic equations and hence is a geodesic. The only way $u' \equiv 0$ on an interval is that γ coincides with a parallel there, which is excluded by hypothesis. Therefore γ is a geodesic.

Remark. For a geodesic on a surface of revolution, $Gv' = \rho \cos \theta$ is constant (Clairaut's relation). This problem shows a partial converse: if a curve satisfies that relation and is not a parallel, it must be a geodesic.

Problem 4: Geodesics on a cone

Let $a > 0$ and consider the half-cone

$$\Sigma = \{(x, y, z) \in \mathbb{R}^3 \mid z^2 = a(x^2 + y^2), z > 0\}.$$

(a) Local isometry and self-intersection

Problem. Show that Σ is locally isometric to the Euclidean plane. By opening the cone into a planar sector, or otherwise, show that when $a = 3$ no geodesic on Σ intersects itself, but for $a > 3$ there exist geodesics which self-intersect.

Solution. Set $r = \sqrt{x^2 + y^2}$, $z = \sqrt{a}r$ and parametrize

$$X(r, \phi) = (r \cos \phi, r \sin \phi, \sqrt{a}r), \quad r > 0, \phi \in \mathbb{R}.$$

Then

$$g = (1 + a) dr^2 + r^2 d\phi^2.$$

Define $s = \sqrt{1 + a}r$. Then $ds^2 = (1 + a) dr^2$ and

$$g = ds^2 + \frac{s^2}{1 + a} d\phi^2 = ds^2 + s^2 d\psi^2, \quad \psi = \frac{\phi}{\sqrt{1 + a}}.$$

This is the Euclidean metric in polar coordinates (s, ψ) . Since ϕ has period 2π , the angle ψ has period

$$\alpha = \frac{2\pi}{\sqrt{1 + a}}.$$

Thus Σ is locally isometric to the Euclidean plane sector

$$S_\alpha = \{(s, \psi) \mid s > 0, 0 < \psi < \alpha\},$$

with the two radial edges identified by rotation through the angle α .

Geodesics on Σ correspond to straight lines in S_α , and two points of S_α represent the same point of Σ if they are related by rotation about the origin through an integer multiple of α .

If $a = 3$ then $\sqrt{1 + a} = 2$ and $\alpha = \pi$. Rotation by π sends a line L to:

- itself, if L passes through the origin (this would correspond to a geodesic through the cone apex, which is not in Σ),
- a parallel disjoint line, if L does not pass through the origin.

Thus for lines corresponding to geodesics on Σ , L and its rotated copy $R_{\pm\pi}(L)$ are disjoint. Hence no geodesic self-intersects when $a = 3$.

If $a > 3$ then $\sqrt{1 + a} > 2$ and $\alpha < \pi$. For any such angle, one can choose a line L not through the origin so that L intersects its rotated copy $R_\alpha(L)$ in a point $p \neq 0$ lying inside the sector S_α . The two points p and $R_{\alpha^{-1}}(p)$ on L correspond to the same point of Σ , so the corresponding geodesic on Σ passes through this point twice and thus self-intersects. Therefore for $a > 3$ there exist self-intersecting geodesics.

(b) Smallest height and Clairaut

Problem. Let γ be a geodesic on Σ which meets the parallel $z = 1$ at angle θ_0 (angle with the parallel). If $\theta_0 \neq \pi/2$, show that the smallest value $z_{\min}$ of z along γ is achieved, and using Clairaut's relation show that $z_{\min}$ is independent of a . What happens when $\theta_0 = \pi/2$?

Solution. The cone is a surface of revolution about the z -axis with radius

$$\rho = r = \frac{z}{\sqrt{a}}.$$

For a geodesic γ on a surface of revolution, Clairaut's relation gives

$$\rho \cos \theta = C,$$

where θ is the angle between γ and the parallels and C is constant along γ .

At the intersection with $z = 1$ we have $\rho = 1/\sqrt{a}$, so

$$C = \frac{1}{\sqrt{a}} \cos \theta_0. \quad (*)$$

If $\theta_0 \neq \pi/2$ then $\cos \theta_0 > 0$, hence $C > 0$. Since $0 \leq \theta \leq \pi/2$ along γ , we have $\cos \theta \leq 1$ and therefore

$$\rho = \frac{C}{\cos \theta} \geq C.$$

Thus ρ has a positive lower bound. In the planar sector picture, γ lifts to a straight line L which has a unique closest point to the origin; at that point the line is tangent to the circle of radius $\rho_{\min}$, so $\theta = 0$ and $\cos \theta = 1$. Hence

$$\rho_{\min} = C.$$

Consequently

$$z_{\min} = \sqrt{a} \rho_{\min} = \sqrt{a} C = \sqrt{a} \cdot \frac{1}{\sqrt{a}} \cos \theta_0 = \cos \theta_0,$$

which is independent of a .

If $\theta_0 = \pi/2$ then $\cos \theta_0 = 0$ and $(*)$ gives $C = 0$, so $\rho \cos \theta \equiv 0$. Since $\cos \theta \geq 0$, this forces $\cos \theta \equiv 0$ and thus $\theta \equiv \pi/2$: the geodesic is a meridian of the cone, everywhere orthogonal to the parallels. Along such a geodesic $z \rightarrow 0$ as one approaches the apex; the infimum of z is 0, but this value is not attained because the apex is not contained in Σ .

Definition 9 (Surface of revolution). *Let the z -axis be fixed in $\mathbb{R}^3$ and let $(\rho(u), 0, z(u))$, $u \in I$, be a smooth plane curve in the rz -plane with $\rho(u) \geq 0$. The surface of revolution obtained by rotating this curve about the z -axis is*

$$\Sigma = \{(\rho(u) \cos v, \rho(u) \sin v, z(u)) \mid u \in I, v \in [0, 2\pi)\}.$$

In the coordinates (u, v) the induced metric has the form

$$g = E(u) du^2 + G(u) dv^2, \quad E(u) = \rho'(u)^2 + z'(u)^2, \quad G(u) = \rho(u)^2.$$

Definition 10 (Angle with parallels). *Let $\gamma(s) = (u(s), v(s))$ be a unit-speed curve on Σ . The unit tangent vector is*

$$T = u'(s) \partial_u + v'(s) \partial_v,$$

and the unit vector tangent to the parallel at the same point is

$$P = \frac{1}{\sqrt{G(u)}} \partial_v.$$

The angle $\theta(s)$ between γ and the parallel is defined by $\cos \theta = g(T, P)$. A short computation gives

$$\cos \theta = \sqrt{G(u)} v'(s) = \rho(u) v'(s).$$

Theorem 1.5 (Clairaut's relation). *Let Σ be a surface of revolution as above and let γ be a unit-speed geodesic on Σ . Then the quantity*

$$G(u(s)) v'(s) = \rho(u(s)) \cos \theta(s)$$

is constant along γ . Equivalently,

$$\boxed{\rho(\gamma(s)) \cos \theta(s) = C.}$$

Example 1.6 (Sphere). On the unit sphere in spherical coordinates (φ, θ) (colatitude and longitude) we have $\rho = \sin \varphi$, so for a geodesic (great circle)

$$\sin \varphi \cos \theta = C.$$

When the geodesic approaches the poles ($\sin \varphi$ small), it becomes more tangent to the parallels ($\theta \rightarrow 0$), and when it crosses the equator ($\varphi = \pi/2$) the angle with the equator is fixed.

Example 1.7 (Cone). For the cone $z^2 = a(x^2 + y^2)$, $z > 0$, we have $\rho = z/\sqrt{a}$, hence a geodesic satisfies

$$\frac{z}{\sqrt{a}} \cos \theta = C.$$

If a geodesic meets the parallel $z = 1$ at angle θ_0 , then $C = \frac{1}{\sqrt{a}} \cos \theta_0$ and the minimal height along the geodesic is $z_{\min} = \cos \theta_0$, independent of a .

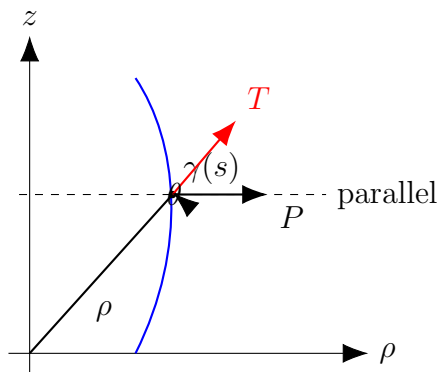


Figure 1: Meridional section of a surface of revolution: ρ is the distance to the axis, θ is the angle between the geodesic and the parallel. Clairaut: $\rho \cos \theta = \text{const}$.

Example: a cone is locally isometric to a planar sector

We explain why a right circular cone (with its tip removed) is a developable surface, i.e. locally isometric to a region of the plane. Geometrically, one can cut the cone along a generator and unroll it into a sector of a circle; the analytic computation below shows that this unrolling preserves the first fundamental form.

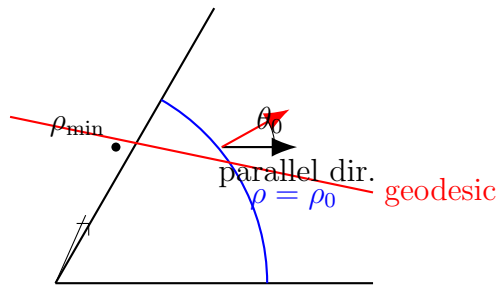


Figure 2: Unrolled cone: geodesics are straight lines in the sector. The distance ρ to the origin has a unique minimum $\rho_{\min}$, where the line is tangent to a circle ($\theta = 0$). Clairaut gives $\rho_{\min} = C$ and hence $z_{\min} = \cos \theta_0$ on the cone.

Geometric picture. Consider a party-hat cone of slant height ρ and base radius R_b . If we cut the cone along one generator and flatten it onto the plane, we obtain a circular sector of radius ρ and some angle Θ . The arc length of the base circle equals the arc length of the boundary of the sector:

$$2\pi R_b = \Theta \cdot \rho,$$

hence $\Theta = 2\pi R_b/\rho < 2\pi$. Gluing the straight sides of this sector reproduces the cone.

The key point is that in this process we only *bend* the surface, without stretching or compressing it. Therefore distances on the cone coincide with distances on the planar sector: the two surfaces are isometric.

Analytic proof. Consider a right circular cone whose tip is at the origin and whose symmetry axis is the z -axis. Fix an angle $\alpha \in (0, \pi/2)$ between any generator and the axis. A convenient parametrisation is

$$X(\rho, \phi) = (\rho \sin \alpha \cos \phi, \rho \sin \alpha \sin \phi, \rho \cos \alpha), \quad \rho > 0, \phi \in (0, 2\pi),$$

where ρ is the *slant distance from the apex* and ϕ is the angle around the axis.

We compute the first fundamental form. First derivatives:

$$X_\rho = (\sin \alpha \cos \phi, \sin \alpha \sin \phi, \cos \alpha),$$

$$X_\phi = (-\rho \sin \alpha \sin \phi, \rho \sin \alpha \cos \phi, 0).$$

Thus

$$E = \langle X_\rho, X_\rho \rangle = \sin^2 \alpha (\cos^2 \phi + \sin^2 \phi) + \cos^2 \alpha = 1,$$

$$F = \langle X_\rho, X_\phi \rangle = 0,$$

$$G = \langle X_\phi, X_\phi \rangle = \rho^2 \sin^2 \alpha (\sin^2 \phi + \cos^2 \phi) = \rho^2 \sin^2 \alpha.$$

Therefore the induced metric on the cone is

$$ds^2 = E d\rho^2 + 2F d\rho d\phi + G d\phi^2 = d\rho^2 + \rho^2 \sin^2 \alpha d\phi^2.$$

Now consider a planar sector with polar coordinates (ρ, θ) , endowed with the usual Euclidean metric

$$ds_{\text{plane}}^2 = d\rho^2 + \rho^2 d\theta^2.$$

Define a map from this sector to the cone by the change of angular variable

$$\theta = \sin \alpha \phi \quad \Longleftrightarrow \quad \phi = \frac{\theta}{\sin \alpha}.$$

Then $d\theta = \sin \alpha d\phi$, and if we pull back the cone metric via this map we obtain

$$d\rho^2 + \rho^2 \sin^2 \alpha d\phi^2 = d\rho^2 + \rho^2 (d\theta)^2 = ds_{\text{plane}}^2.$$

Thus the map

$$(\rho, \theta) \longmapsto (\rho \sin \alpha \cos(\theta / \sin \alpha), \rho \sin \alpha \sin(\theta / \sin \alpha), \rho \cos \alpha)$$

is an *isometry* between a planar sector

$$\{(\rho, \theta) \mid 0 < \rho < \infty, 0 < \theta < \Theta\}$$

(with $\Theta = 2\pi \frac{R_b}{\rho}$ as above) and the cone with its apex removed.

Distances and angles are preserved because the first fundamental form in these coordinates coincides exactly with the Euclidean metric on the sector. Consequently the cone (without the tip) is locally isometric to a region of the plane and hence is a *developable* surface (its Gaussian curvature satisfies $K \equiv 0$ away from the vertex).

Remark 1.8. Globally, the cone is not isometric to the whole plane, but only to a sector of angle $\Theta < 2\pi$. However, in small neighbourhoods that do not wrap around the apex, the cone and the plane are indistinguishable from the metric point of view. This illustrates the local nature of developability.

Theory: Surfaces of revolution

A *surface of revolution* is obtained by rotating a plane curve around a fixed line (the axis of revolution). We describe the standard case where the axis is the z -axis.

Generating curve. Let

$$\gamma(u) = (\rho(u), 0, z(u)), \quad u \in I \subset \mathbb{R},$$

be a smooth curve in the xz -plane (equivalently, in the (ρ, z) half-plane), where

$$\rho(u) \geq 0 \quad \text{is the distance from the } z\text{-axis}, \quad z(u) \in \mathbb{R} \quad \text{is the height.}$$

This curve is called the *profile* or *meridian* of the surface. The line of revolution is the z -axis.

Parametrisation of the surface of revolution. Rotating the point $\gamma(u)$ by an angle $v \in [0, 2\pi)$ around the z -axis gives the point

$$(\rho(u) \cos v, \rho(u) \sin v, z(u)).$$

Thus a parametrisation of the surface of revolution generated by γ is

$$X : I \times [0, 2\pi) \rightarrow \mathbb{R}^3, \quad X(u, v) = (\rho(u) \cos v, \rho(u) \sin v, z(u)).$$

Here

- u moves along the generating curve (meridians),
- v is the rotation angle around the z -axis (parallels).

First fundamental form. We compute the partial derivatives:

$$X_u = (\rho'(u) \cos v, \rho'(u) \sin v, z'(u)),$$

$$X_v = (-\rho(u) \sin v, \rho(u) \cos v, 0).$$

The coefficients of the first fundamental form are

$$E(u) = \langle X_u, X_u \rangle = \rho'(u)^2(\cos^2 v + \sin^2 v) + z'(u)^2 = \rho'(u)^2 + z'(u)^2,$$

$$F(u) = \langle X_u, X_v \rangle = 0 \quad (\text{orthogonal coordinates}),$$

$$G(u) = \langle X_v, X_v \rangle = \rho(u)^2(\sin^2 v + \cos^2 v) = \rho(u)^2.$$

Thus the induced metric (first fundamental form) is

$$g = E(u) du^2 + 2F(u) du dv + G(u) dv^2 = E(u) du^2 + G(u) dv^2,$$

with

$$E(u) = \rho'(u)^2 + z'(u)^2, \quad G(u) = \rho(u)^2.$$

Examples of surfaces of revolution

1. **Cylinder of radius R .** Take $\rho(u) = R$ and $z(u) = u$:

$$X(u, v) = (R \cos v, R \sin v, u), \quad E(u) = 1, \quad G(u) = R^2.$$

2. **Right circular cone.** Take $\rho(u) = au$, $z(u) = bu$ with $a, b > 0$:

$$X(u, v) = (au \cos v, au \sin v, bu), \quad E(u) = a^2 + b^2, \quad G(u) = a^2 u^2.$$

3. **Sphere of radius R .** Using the polar angle $u \in (0, \pi)$ as parameter for the meridian,

$$\rho(u) = R \sin u, \quad z(u) = R \cos u,$$

so

$$X(u, v) = (R \sin u \cos v, R \sin u \sin v, R \cos u),$$

and

$$E(u) = R^2, \quad G(u) = R^2 \sin^2 u.$$

Problem 5 – A surface with no minimizing geodesic between two points

Statement. Give an example of a connected smooth surface $\Sigma \subset \mathbb{R}^3$ and points $p, q \in \Sigma$ for which the infimum

$$\inf_{\gamma} L(\gamma),$$

taken over all piecewise smooth curves $\gamma : [a, b] \rightarrow \Sigma$ with $\gamma(a) = p$ and $\gamma(b) = q$, is strictly smaller than the length of any piecewise smooth curve γ between p and q .

Example and solution. Let

$$\Sigma = \{(x, y, 0) \in \mathbb{R}^3 : (x, y) \neq (0, 0)\},$$

the plane $z = 0$ with the origin removed. This is a connected smooth surface in $\mathbb{R}^3$.

Choose

$$p = (-1, 0, 0), \quad q = (1, 0, 0).$$

In the full plane, the straight line segment from p to q has length 2 and passes through the origin. Since the origin is removed, this segment is not contained in Σ and is not an admissible curve.

Any piecewise smooth curve $\gamma \subset \Sigma$ joining p to q must go around the origin, and so has length strictly greater than 2. On the other hand, for each $\varepsilon > 0$ we can construct a curve that avoids the origin but has length arbitrarily close to 2; for instance, in the xy -plane:

- the line segment from $(-1, 0)$ to $(-\varepsilon, 0)$,
- a semicircle of radius ε around the origin,
- the line segment from $(\varepsilon, 0)$ to $(1, 0)$.

Lifting to $\Sigma \subset \mathbb{R}^3$ we obtain curves whose lengths tend to 2 as $\varepsilon \rightarrow 0$, but none of which have length exactly 2.

Therefore

$$\inf_{\gamma} L(\gamma) = 2$$

but no curve in Σ connecting p and q attains this infimum. Hence the infimum of lengths is strictly smaller than the length of any admissible curve: there is no minimizing geodesic from p to q on Σ .

Problem 6 – A ruled surface along the binormal; η is geodesic

Statement. Let $\eta : [s_0, s_1] \rightarrow \mathbb{R}^3$ be an embedded smooth curve parametrised by arc-length, with $\eta''(s) \neq 0$ for all s (nonzero curvature). The *binormal vector* to η is the unit vector

$$b(s) = \frac{\eta'(s) \times \eta''(s)}{\|\eta'(s) \times \eta''(s)\|}.$$

Consider the ruled surface with parametrisation

$$\sigma(u, v) = \eta(u) + v b(u), \quad u \in (s_0, s_1), \quad -\varepsilon < v < \varepsilon, \quad \varepsilon > 0.$$

- (i) Show that $D\sigma|_p$ is injective for $p \in \eta$, and hence is injective for all points in the image of σ with $|v|$ sufficiently small.
- (ii) Deduce that, if ε is sufficiently small, the parametrisation σ is an injective map and that the image of σ defines a smooth surface in $\mathbb{R}^3$.
- (iii) Show that η is a geodesic on the resulting surface.

(i) Differential at points on η . We compute

$$\sigma_u(u, v) = \eta'(u) + v b'(u), \quad \sigma_v(u, v) = b(u).$$

For points on η we have $v = 0$, so

$$\sigma_u(u, 0) = \eta'(u), \quad \sigma_v(u, 0) = b(u).$$

Since η is parametrised by arc-length, $\eta'(u)$ is the unit tangent $t(u)$. Because $\eta''(s) \neq 0$, the Frenet frame $(t(u), n(u), b(u))$ is well-defined; in particular $b(u)$ is a unit vector orthogonal to $t(u)$, so $t(u)$ and $b(u)$ are nonzero and linearly independent.

Hence the columns $\sigma_u(u, 0)$ and $\sigma_v(u, 0)$ are linearly independent for each u , and the differential

$$D\sigma_{(u,0)} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

has rank 2. By continuity of σ_u and σ_v , this remains true for all points with $|v|$ sufficiently small, so $D\sigma$ is injective on a thin strip around the curve.

(ii) Injectivity and smooth surface. Since η is an embedded curve and $[s_0, s_1]$ is compact, the tubular neighbourhood theorem implies that there exists $r > 0$ such that the map

$$(u, v, w) \mapsto \eta(u) + v n(u) + w b(u)$$

is an embedding from $(s_0, s_1) \times (-r, r)^2$ into $\mathbb{R}^3$. In particular, restricting to the binormal direction (say $v = 0$) and to $|w| < r$ shows that

$$(u, w) \mapsto \eta(u) + w b(u)$$

is injective. Choosing $0 < \varepsilon < r$ and writing v in place of w we obtain that

$$\sigma(u, v) = \eta(u) + v b(u), \quad u \in (s_0, s_1), \quad |v| < \varepsilon,$$

is an injective immersion. Its image is therefore a smooth surface (an embedded 2-dimensional submanifold) in $\mathbb{R}^3$.

(iii) η is a geodesic on the surface. Along the curve η in the surface we have $v = 0$ and

$$\sigma(u, 0) = \eta(u), \quad \sigma_u(u, 0) = \eta'(u) = t(u), \quad \sigma_v(u, 0) = b(u).$$

The surface normal along η is

$$N(u) = \frac{\sigma_u(u, 0) \times \sigma_v(u, 0)}{\|\sigma_u(u, 0) \times \sigma_v(u, 0)\|} = \frac{t(u) \times b(u)}{\|t(u) \times b(u)\|}.$$

In the Frenet frame $(t(u), n(u), b(u))$ we have $t(u) \times n(u) = b(u)$; thus $t(u) \times b(u) = -n(u)$, and

$$N(u) = \pm n(u).$$

The curvature vector of η is

$$\eta''(u) = \kappa(u) n(u),$$

where $\kappa(u) > 0$ is the curvature. Therefore $\eta''(u)$ is parallel to $N(u)$, i.e. it is normal to the surface. The tangential component of $\eta''(u)$ in the surface is zero, so the geodesic curvature of η on this surface vanishes. Hence η is a geodesic on the ruled surface $\sigma(u, v)$.

Problem 6 – Charts

Chart 1 – Data and parametrisation.

- Curve: $\eta : [s_0, s_1] \rightarrow \mathbb{R}^3$ smooth, embedded, arc-length parametrised:

$$\|\eta'(s)\| = 1, \quad \eta''(s) \neq 0.$$

- Frenet frame:

$$t(s) = \eta'(s), \quad n(s) = \frac{\eta''(s)}{\|\eta''(s)\|}, \quad b(s) = \frac{t(s) \times n(s)}{\|t(s) \times n(s)\|},$$

so (t, n, b) is an orthonormal basis along the curve.

- Ruled surface along the binormal:

$$\sigma(u, v) = \eta(u) + v b(u), \quad u \in (s_0, s_1), \quad |v| < \varepsilon.$$

Chart 2 – Differential and the binormal strip.

- First derivatives:

$$\sigma_u(u, v) = \eta'(u) + v b'(u), \quad \sigma_v(u, v) = b(u).$$

- Along the curve η (i.e. at $v = 0$):

$$\sigma_u(u, 0) = t(u), \quad \sigma_v(u, 0) = b(u).$$

Here $t(u)$ and $b(u)$ are unit, perpendicular, nonzero; hence they are linearly independent and

$$\text{rank } D\sigma_{(u,0)} = 2.$$

- By continuity of σ_u and σ_v , for all (u, v) with $|v|$ sufficiently small, $\sigma_u(u, v)$ and $\sigma_v(u, v)$ remain independent. Thus σ is an immersion on $(s_0, s_1) \times (-\varepsilon, \varepsilon)$.
- Using that η is embedded and $[s_0, s_1]$ is compact, the tubular neighbourhood theorem implies that, for $\varepsilon > 0$ small enough, σ is also injective. Its image is then a smooth embedded surface (a thin binormal strip) in $\mathbb{R}^3$.

Chart 3 – Why η is a geodesic on the surface.

- On the surface along η we have $v = 0$ and

$$\sigma(u, 0) = \eta(u), \quad \sigma_u(u, 0) = t(u), \quad \sigma_v(u, 0) = b(u).$$

- The surface normal is

$$N(u) = \frac{\sigma_u(u, 0) \times \sigma_v(u, 0)}{\|\sigma_u(u, 0) \times \sigma_v(u, 0)\|} = \frac{t(u) \times b(u)}{\|t(u) \times b(u)\|} = \pm n(u),$$

since in the Frenet frame (t, n, b) we have $t \times n = b$ and hence $t \times b = -n$.

- The curvature vector of η is

$$\eta''(u) = \kappa(u) n(u),$$

with $\kappa(u) > 0$. Thus $\eta''(u)$ is parallel to $N(u)$, i.e. purely normal to the surface.

- Therefore the tangential component of $\eta''(u)$ within the surface vanishes, so the geodesic curvature of η on this surface is zero. By the intrinsic characterisation of geodesics, η is a geodesic on the ruled surface $\sigma(u, v)$.

Problem 7: Isometries and Möbius maps on the 2-sphere

Statement. Let $S^2 \subset \mathbb{R}^3$ be the unit sphere with the standard round metric.

1. Let $f : S^2 \rightarrow S^2$ be a diffeomorphism which is also a global isometry. Using that f sends geodesics to geodesics (or otherwise), show that f is the restriction to S^2 of an element of the orthogonal group $O(3)$.
2. Identifying $\mathbb{C} \cup \{\infty\}$ with $S^2 \subset \mathbb{R}^3$ via stereographic projection, prove that the Möbius group $\text{PSL}(2, \mathbb{C})$ acts on the abstract smooth surface S^2 by diffeomorphisms.
Hint: Check that the generators $z \mapsto az + b$ and $z \mapsto 1/z$ of the Möbius group act smoothly from the original definition of smooth maps in terms of a smooth atlas.
3. If a Möbius map A defines an isometry of S^2 , show that it commutes with the antipodal map $a : S^2 \rightarrow S^2$ given by $a(x, y, z) = (-x, -y, -z)$.

Theory / Background.

- The isometry group of the round sphere (S^2, g_{round}) is the group of restrictions of orthogonal transformations $A \in O(3)$. Great circles (intersections of S^2 with 2-planes through the origin) are exactly the geodesics.
- For $p \in S^2$ the exponential map

$$\exp_p : T_p S^2 \rightarrow S^2, \quad \exp_p(v) = \cos |v| p + \sin |v| \frac{v}{|v|},$$

is a diffeomorphism from $\{v \in T_p S^2 : |v| < \pi\}$ onto $S^2 \setminus \{-p\}$.

- Stereographic projection $\sigma : S^2 \setminus \{N\} \rightarrow \mathbb{C}$ from the north pole N is a smooth diffeomorphism and extends to a homeomorphism $\sigma : S^2 \rightarrow \hat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$, $\sigma(N) = \infty$.
- Möbius transformations are maps

$$z \mapsto \frac{az + b}{cz + d}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{C}),$$

considered up to nonzero scalar multiples; they form the group $\text{PSL}(2, \mathbb{C})$. The group is generated by translations $z \mapsto z + b$, dilations/rotations $z \mapsto az$, and inversion $z \mapsto 1/z$.

- Any linear map $B : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ commutes with the antipodal map $-\text{id}$: $B(-x) = -B(x)$.

Solution.

(a) Isometries of S^2 come from $O(3)$. Let $f : S^2 \rightarrow S^2$ be an isometry.

Step 1: Extend the derivative. Choose $p \in S^2$ and set $q = f(p)$. Because f is an isometry, its differential

$$df_p : T_p S^2 \rightarrow T_q S^2$$

is an isometry of inner-product spaces.

Choose orthonormal bases $\{e_1, e_2\}$ of $T_p S^2$ and $\{e'_1, e'_2\}$ of $T_q S^2$ such that $e'_i = df_p(e_i)$. Then $\{e_1, e_2, p\}$ and $\{e'_1, e'_2, q\}$ are orthonormal bases of $\mathbb{R}^3$. Define $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by

$$A(e_i) = e'_i, \quad A(p) = q.$$

In these bases the matrix of A is orthogonal, so $A \in O(3)$. In particular $A(S^2) = S^2$ and A restricts to an isometry of S^2 .

Step 2: Reduce to an isometry fixing a point with identity derivative. Define

$$g := A^{-1} \circ f : S^2 \rightarrow S^2.$$

Then g is an isometry, $g(p) = p$, and

$$dg_p = (A^{-1})_* \circ df_p = \text{id}_{T_p S^2}.$$

Step 3: Show that $g = \text{id}_{S^2}$. Let $v \in T_p S^2$ with $|v| < \pi$. Let

$$\gamma_v(t) := \exp_p(tv)$$

be the unique geodesic with $\gamma_v(0) = p$, $\gamma'_v(0) = v$. Since g is an isometry and $g(p) = p$, the curve $g \circ \gamma_v$ is a geodesic starting at p . Its initial velocity is

$$(g \circ \gamma_v)'(0) = dg_p(\gamma'_v(0)) = dg_p(v) = v,$$

so $g \circ \gamma_v$ and γ_v are geodesics with the same initial data. By uniqueness of geodesics, $g(\gamma_v(t)) = \gamma_v(t)$ for all t with $|t| < \pi$.

The exponential map $\exp_p$ is a diffeomorphism from $\{v \in T_p S^2 : |v| < \pi\}$ onto $S^2 \setminus \{-p\}$, so

$$g(x) = x \quad \text{for all } x \in S^2 \setminus \{-p\}.$$

By continuity, $g(-p) = \lim_{t \rightarrow \pi^-} g(\gamma_v(t)) = \lim_{t \rightarrow \pi^-} \gamma_v(t) = -p$, hence g is the identity on all of S^2 .

Step 4: Conclusion. We have $f = A \circ g = A|_{S^2}$, so f is the restriction to S^2 of some $A \in O(3)$.

(b) The Möbius group acts smoothly on S^2 . Let $\sigma : S^2 \rightarrow \hat{\mathbb{C}}$ be stereographic projection (extended by $\sigma(N) = \infty$). For $M \in \text{PSL}(2, \mathbb{C})$ define

$$\Phi_M := \sigma^{-1} \circ M \circ \sigma : S^2 \rightarrow S^2.$$

Group action. Since σ and M are bijections, so is Φ_M . For $M_1, M_2 \in \text{PSL}(2, \mathbb{C})$ we have

$$\Phi_{M_1 M_2} = \sigma^{-1} \circ M_1 M_2 \circ \sigma = (\sigma^{-1} \circ M_1 \circ \sigma) \circ (\sigma^{-1} \circ M_2 \circ \sigma) = \Phi_{M_1} \circ \Phi_{M_2},$$

and clearly $\Phi_{\text{id}} = \text{id}_{S^2}$. Thus $\text{PSL}(2, \mathbb{C})$ acts on S^2 .

Smoothness. Consider the atlas $\{(U_0, \varphi_0), (U_1, \varphi_1)\}$ on S^2 , where

$$U_0 = S^2 \setminus \{N\}, \quad \varphi_0 = \sigma, \quad U_1 = S^2 \setminus \{S\}, \quad \varphi_1 = z \mapsto \frac{1}{\sigma(z)}.$$

A map $\Psi : S^2 \rightarrow S^2$ is smooth iff each coordinate representation $\varphi_i \circ \Psi \circ \varphi_j^{-1}$ is smooth between open subsets of $\mathbb{C}$.

The group $\text{PSL}(2, \mathbb{C})$ is generated by

$$T_{a,b} : z \mapsto az + b, \quad I : z \mapsto 1/z.$$

These maps are rational functions of z , hence smooth on the chart U_0 . On the chart near ∞ (where we use coordinates $w = 1/z$) their coordinate expressions remain rational (e.g. $w \mapsto w/(a + bw)$ for $z \mapsto az + b$), hence smooth. Therefore each generator induces a smooth diffeomorphism of S^2 .

Compositions and inverses of smooth diffeomorphisms are smooth diffeomorphisms. Hence for every $M \in \text{PSL}(2, \mathbb{C})$ the map Φ_M is a diffeomorphism. Thus $\text{PSL}(2, \mathbb{C})$ acts on S^2 by diffeomorphisms.

(c) Isometric Möbius maps commute with the antipodal map. Suppose a Möbius map A induces an isometry of (S^2, g_{round}) . By part (b) this gives a diffeomorphism $f_A : S^2 \rightarrow S^2$. Since f_A is an isometry, part (a) implies that there exists $B \in O(3)$ with

$$f_A = B|_{S^2}.$$

Let $a : S^2 \rightarrow S^2$ be the antipodal map $a(x) = -x$. Because B is linear, $B(-x) = -B(x)$ for all $x \in \mathbb{R}^3$. Hence, for every $x \in S^2$,

$$f_A(a(x)) = B(-x) = -B(x) = a(B(x)) = a(f_A(x)).$$

Thus $f_A \circ a = a \circ f_A$, i.e. the isometric Möbius map commutes with the antipodal map.

Examples.

1. For $\theta \in \mathbb{R}$, the rotation

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \in SO(3)$$

restricts to an isometry of S^2 . In stereographic coordinates $z = x + iy$ this corresponds to the Möbius map $z \mapsto e^{i\theta} z$.

2. The dilation $z \mapsto 2z$ is a Möbius transformation but does not come from any $B \in O(3)$, hence its induced map on S^2 is not an isometry of the round metric.
3. Every element of $SO(3)$ corresponds, via stereographic projection, to a Möbius map of the form

$$z \mapsto \frac{\alpha z + \beta}{-\bar{\beta} z + \bar{\alpha}}, \quad |\alpha|^2 + |\beta|^2 = 1,$$

and these are precisely the orientation-preserving isometries of S^2 .

Möbius transformations, $\mathrm{PSL}(2, \mathbb{C})$, and the action on S^2

In this subsection we explain in detail why the Möbius group can be identified with $\mathrm{PSL}(2, \mathbb{C})$ and why, after identifying $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ with $S^2 \subset \mathbb{R}^3$ via stereographic projection, this group acts on the abstract smooth surface S^2 by diffeomorphisms.

1. The Riemann sphere and stereographic projection

We consider the unit sphere

$$S^2 = \{(x, y, t) \in \mathbb{R}^3 : x^2 + y^2 + t^2 = 1\}.$$

Let $N = (0, 0, 1)$ be the north pole, and let

$$\Pi : S^2 \setminus \{N\} \longrightarrow \mathbb{R}^2 \cong \mathbb{C}$$

be the stereographic projection from N onto the plane $\{t = 0\}$. In coordinates one checks that

$$\Pi(x, y, t) = \left(\frac{x}{1-t}, \frac{y}{1-t} \right),$$

so if we identify $\mathbb{R}^2$ with $\mathbb{C}$ via $(X, Y) \mapsto X + iY$, then

$$\Pi(x, y, t) = z = \frac{x}{1-t} + i \frac{y}{1-t}.$$

We extend Π to a bijection

$$\Phi : S^2 \longrightarrow \widehat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$$

by setting $\Phi(N) = \infty$ and $\Phi|_{S^2 \setminus \{N\}} = \Pi$. This is a homeomorphism, and in fact a smooth diffeomorphism in the sense of smooth manifolds if we equip $\widehat{\mathbb{C}}$ with the standard structure of a 1-dimensional complex manifold (the Riemann sphere). Its inverse $\Phi^{-1} : \widehat{\mathbb{C}} \rightarrow S^2$ is given by the classical formulas (for $z = x + iy \in \mathbb{C}$)

$$\Phi^{-1}(z) = \left(\frac{2x}{|z|^2 + 1}, \frac{2y}{|z|^2 + 1}, \frac{|z|^2 - 1}{|z|^2 + 1} \right), \quad \Phi^{-1}(\infty) = N.$$

Thus S^2 and $\widehat{\mathbb{C}}$ are the same smooth surface, viewed in two different ways; we will freely pass between them via Φ .

2. Möbius transformations on the extended complex plane

A *Möbius transformation* (or linear fractional transformation) is a map $f : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ of the form

$$f(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc \neq 0,$$

with the usual conventions

$$f\left(-\frac{d}{c}\right) = \infty \quad (c \neq 0), \quad f(\infty) = \begin{cases} \frac{a}{c}, & c \neq 0, \\ \infty, & c = 0. \end{cases}$$

It is well known (and easy to check) that:

- Möbius transformations form a group under composition;
- the map $z \mapsto z$ is the identity element;
- every Möbius transformation is a homeomorphism of $\widehat{\mathbb{C}}$ with continuous inverse of the same form.

3. Möbius transformations and 2×2 matrices

To each Möbius transformation $f(z) = \frac{az + b}{cz + d}$ we associate the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{C}).$$

Composition of Möbius transformations corresponds to matrix multiplication: if

$$f_A(z) = \frac{az + b}{cz + d}, \quad f_B(z) = \frac{\alpha z + \beta}{\gamma z + \delta},$$

then for all $z \in \widehat{\mathbb{C}}$ for which both sides are defined we have

$$(f_B \circ f_A)(z) = f_{BA}(z),$$

where BA is the ordinary product of matrices.

However, the matrix representation is not unique: multiplying the matrix by a nonzero scalar does not change the Möbius transformation. Indeed, if $\lambda \in \mathbb{C}^\times$, then

$$f_{\lambda A}(z) = \frac{(\lambda a)z + \lambda b}{(\lambda c)z + \lambda d} = \frac{az + b}{cz + d} = f_A(z).$$

Thus the same Möbius map corresponds to the whole 1-dimensional family of matrices λA , $\lambda \neq 0$.

We remove this redundancy by working projectively. First, note that we may restrict attention to matrices with determinant 1: if $A \in \mathrm{GL}(2, \mathbb{C})$ has $\det A \neq 0$, then we may rescale A to λA with $\lambda^2 \det A = 1$, so that $\lambda A \in \mathrm{SL}(2, \mathbb{C})$. This rescaling does not change f_A , so every Möbius transformation is represented by *some* matrix in $\mathrm{SL}(2, \mathbb{C})$.

Moreover, within $\mathrm{SL}(2, \mathbb{C})$ the only nontrivial scalar multiple that preserves determinant 1 is $-I$, since if $\lambda A \in \mathrm{SL}(2, \mathbb{C})$ with $A \in \mathrm{SL}(2, \mathbb{C})$, then

$$1 = \det(\lambda A) = \lambda^2 \det A = \lambda^2,$$

so $\lambda = \pm 1$. Thus the scalar redundancy inside $\mathrm{SL}(2, \mathbb{C})$ is precisely the central subgroup $\{\pm I\}$.

We therefore define the *projective special linear group*

$$\mathrm{PSL}(2, \mathbb{C}) := \mathrm{SL}(2, \mathbb{C}) / \{\pm I\}.$$

An element of $\mathrm{PSL}(2, \mathbb{C})$ is an equivalence class $[A]$ of matrices $A \in \mathrm{SL}(2, \mathbb{C})$, where $A \sim -A$.

The assignment

$$[A] \mapsto f_A$$

is a well-defined group homomorphism

$$\mathrm{PSL}(2, \mathbb{C}) \longrightarrow \{\text{Möbius transformations on } \widehat{\mathbb{C}}\},$$

which is bijective. Hence we have a natural group isomorphism

$$\mathrm{PSL}(2, \mathbb{C}) \cong \{\text{all Möbius transformations on } \widehat{\mathbb{C}}\}.$$

This is the precise sense in which “the Möbius group is $\mathrm{PSL}(2, \mathbb{C})$ ”.

From the complex-geometric viewpoint, $\widehat{\mathbb{C}}$ is the complex projective line $\mathbb{CP}^1$ and $\mathrm{PSL}(2, \mathbb{C})$ is the group of holomorphic automorphisms of $\mathbb{CP}^1$.

4. Generators of the Möbius group

It is a standard fact that the Möbius group is generated by the following simple maps:

$$\begin{aligned} T_b(z) &= z + b, & b \in \mathbb{C} & \text{ (translations),} \\ D_a(z) &= az, & a \in \mathbb{C}^\times & \text{ (dilations/rotations),} \\ I(z) &= \frac{1}{z} & & \text{ (inversion).} \end{aligned}$$

More precisely, any Möbius transformation $f(z) = \frac{az + b}{cz + d}$ with $ad - bc \neq 0$ can be written as a finite composition of maps of the form T_b , D_a , and I . For example, when $c \neq 0$ one checks that

$$\frac{az + b}{cz + d} = D_{a/c} \circ T_{d/c} \circ I \circ T_{b/a}(z),$$

and similar decompositions hold in all cases. Thus, to understand the smoothness properties of general Möbius maps, it suffices to understand these generators.

5. Induced maps on the sphere via stereographic projection

Given any Möbius transformation $f : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$, we obtain a map

$$F := \Phi^{-1} \circ f \circ \Phi : S^2 \longrightarrow S^2.$$

The exercise asks us to show that F is a *diffeomorphism* of the smooth surface S^2 . Since Φ and Φ^{-1} are already smooth diffeomorphisms, it is enough to check that f is smooth with smooth inverse in the atlas on $\widehat{\mathbb{C}}$ coming from stereographic coordinates.

The important observation is that smoothness of F can be verified in local coordinates. On the chart

$$U = S^2 \setminus \{N\}, \quad \Phi : U \rightarrow \mathbb{C},$$

the coordinate expression of F is

$$\Phi \circ F \circ \Phi^{-1} = f : \mathbb{C} \rightarrow \mathbb{C},$$

which is clearly smooth wherever it is defined (it is a rational function with denominator nonzero). Near the point corresponding to ∞ we switch to another stereographic chart; in those coordinates the expression for f is again rational, hence smooth, and the chart transition maps between the two stereographic charts are themselves Möbius transformations (and therefore smooth). This shows that F is smooth in the sense of smooth manifolds.

Because Möbius transformations form a group under composition and inversion, f has a Möbius inverse f^{-1} , and therefore

$$F^{-1} = \Phi^{-1} \circ f^{-1} \circ \Phi : S^2 \rightarrow S^2$$

is also smooth. Consequently F is a diffeomorphism.

To make this completely concrete it is often convenient to verify the claim first for the generators T_b , D_a and I , and then conclude for all Möbius maps by composition. We briefly spell this out.

Translations and dilations/rotations. Let $f(z) = az + b$ with $a \neq 0$. Then on the chart U ,

$$\Phi \circ F \circ \Phi^{-1}(z) = f(z) = az + b,$$

which is entire and hence smooth. On the overlap with the second stereographic chart (from the south pole) the coordinate expression of F is the composition of f with the smooth chart transition maps; it is therefore smooth as well. The inverse map is represented by the affine map $f^{-1}(z) = a^{-1}(z - b)$, so F is a diffeomorphism.

Inversion. Let $f(z) = 1/z$, extended by $f(0) = \infty$, $f(\infty) = 0$. On $\mathbb{C}^\times$ this is smooth. In the stereographic chart U away from the point $\Phi^{-1}(0)$ (the point corresponding to $z = 0$) the map

$$\Phi \circ F \circ \Phi^{-1}(z) = \frac{1}{z}$$

is smooth. Near $z = \infty$ we switch to the other stereographic chart, where the coordinate $w = 1/z$ is finite; in that chart f is simply $w \mapsto w$, so again we have a smooth expression. The inverse of f is itself ($f^{-1} = f$), so F is a diffeomorphism of S^2 .

Since translation, dilation/rotation, and inversion each induce smooth diffeomorphisms of S^2 via

$$F = \Phi^{-1} \circ f \circ \Phi,$$

and every Möbius transformation is a finite composition of these generators, we conclude:

For every $[A] \in \text{PSL}(2, \mathbb{C})$ the corresponding Möbius transformation f_A determines a diffeomorphism

$$F_A = \Phi^{-1} \circ f_A \circ \Phi : S^2 \rightarrow S^2.$$

The assignment $[A] \mapsto F_A$ defines a smooth action of $\text{PSL}(2, \mathbb{C})$ on the abstract smooth surface S^2 by diffeomorphisms.

Thus, identifying $\widehat{\mathbb{C}}$ with S^2 via stereographic projection, the Möbius group $\text{PSL}(2, \mathbb{C})$ acts on S^2 smoothly and by diffeomorphisms, as required in part (b) of the problem.

Problem 8: Exotic metrics on the disc and on the square

(a) **Statement.** On the open unit disc

$$B(0, 1) = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 < 1\}$$

define an abstract Riemannian metric by

$$g = \frac{du^2 + dv^2}{1 - u^2 - v^2}.$$

Prove directly that diameter segments (straight line segments through the origin) are length-minimizing curves between their endpoints. Show that distances with respect to this metric are bounded, but areas can be unbounded.

(a) Solution. In polar coordinates $u = r \cos \theta$, $v = r \sin \theta$ we have

$$du^2 + dv^2 = dr^2 + r^2 d\theta^2, \quad 1 - u^2 - v^2 = 1 - r^2,$$

so

$$g = \frac{dr^2 + r^2 d\theta^2}{1 - r^2}.$$

Diameters are length-minimizing. Let $\gamma(t) = (r(t), \theta(t))$ be a C^1 curve in $B(0, 1)$. Its speed is

$$|\gamma'(t)|_g = \sqrt{\frac{r'(t)^2 + r(t)^2 \theta'(t)^2}{1 - r(t)^2}} \geq \frac{|r'(t)|}{\sqrt{1 - r(t)^2}}.$$

Hence

$$L_g(\gamma) = \int |\gamma'(t)|_g dt \geq \int \frac{|r'(t)|}{\sqrt{1 - r(t)^2}} dt.$$

Suppose γ joins two points on a fixed diameter, i.e. with polar coordinates (r_1, θ_0) and (r_2, θ_0) . Reparametrising, we may assume that $r(t)$ is monotone. Then

$$\int \frac{|r'(t)|}{\sqrt{1 - r(t)^2}} dt = \int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - r^2}} = |\arcsin r_2 - \arcsin r_1|.$$

Let δ be the diameter segment $\delta(t) = (r(t), \theta_0)$ joining these points. For δ we have $\theta'(t) \equiv 0$, so

$$L_g(\delta) = \int \sqrt{\frac{r'(t)^2}{1 - r(t)^2}} dt = \int_{r_1}^{r_2} \frac{dr}{\sqrt{1 - r^2}} = |\arcsin r_2 - \arcsin r_1|.$$

Therefore $L_g(\gamma) \geq L_g(\delta)$, with equality if and only if $\theta'(t) \equiv 0$, i.e. if γ is contained in the same diameter. Hence diameter segments are length-minimizing curves between their endpoints.

Distances are bounded. For a point of radius r the distance from the origin along a radius is

$$d_g(0, (r, \theta)) = \int_0^r \frac{ds}{\sqrt{1 - s^2}} = \arcsin r.$$

Thus for any $p, q \in B(0, 1)$ with radii r_p, r_q we have

$$d_g(p, q) \leq d_g(p, 0) + d_g(0, q) = \arcsin r_p + \arcsin r_q < \pi.$$

So the metric space $(B(0, 1), d_g)$ has finite diameter.

Areas can be unbounded. The metric is conformal to the Euclidean metric: $g = \lambda(du^2 + dv^2)$ with

$$\lambda(u, v) = \frac{1}{1 - u^2 - v^2}.$$

Hence the area element is

$$dA_g = \lambda du dv = \frac{du dv}{1 - u^2 - v^2}.$$

In polar coordinates the area of the Euclidean disc of radius r is

$$\text{Area}_g(B(0, r)) = \int_0^{2\pi} \int_0^r \frac{\rho}{1 - \rho^2} d\rho d\theta = 2\pi \int_0^r \frac{\rho}{1 - \rho^2} d\rho.$$

With $t = \rho^2$,

$$\text{Area}_g(B(0, r)) = \pi \int_0^{r^2} \frac{dt}{1 - t} = -\pi \ln(1 - r^2).$$

As $r \rightarrow 1^-$, the quantity $-\ln(1 - r^2)$ tends to $+\infty$. Thus the areas of such discs become arbitrarily large, so areas in this metric are unbounded.

(b) Statement. Let $V = \{(u, v) \in \mathbb{R}^2 : |u| < 1, |v| < 1\}$ and define

$$g_1 = \frac{du^2}{(1 - u^2)^2} + \frac{dv^2}{(1 - v^2)^2}, \quad g_2 = \frac{du^2}{(1 - v^2)^2} + \frac{dv^2}{(1 - u^2)^2}.$$

1. A map $\gamma : [0, \infty) \rightarrow V$ is a *properly embedded path* if, for every compact $K \subset V$, the preimage $\gamma^{-1}(K)$ is compact (equivalently, for every compact K there is T such that $\gamma(t) \notin K$ for all $t \geq T$). Show that any homeomorphism $h : V \rightarrow V$ sends properly embedded paths to properly embedded paths.
2. Prove that the Riemannian surfaces (V, g_1) and (V, g_2) are not isometric, but that there is an area-preserving diffeomorphism between them.

Hint: Show that exactly one of the two metrics admits a properly embedded path of finite length.

(b)(i) Solution. Let $\gamma : [0, \infty) \rightarrow V$ be a properly embedded path and $h : V \rightarrow V$ a homeomorphism. For any compact $K \subset V$, $h^{-1}(K)$ is compact because h is a homeomorphism. Properness of γ implies that $\gamma^{-1}(h^{-1}(K))$ is compact in $[0, \infty)$. But

$$(h \circ \gamma)^{-1}(K) = \gamma^{-1}(h^{-1}(K)),$$

so $(h \circ \gamma)^{-1}(K)$ is compact. Hence $h \circ \gamma$ is properly embedded.

(b)(ii) Solution. *Area-preserving diffeomorphism.* For the metric g_1 we have

$$E_1(u, v) = \frac{1}{(1 - u^2)^2}, \quad G_1(u, v) = \frac{1}{(1 - v^2)^2},$$

so its area element is

$$dA_{g_1} = \sqrt{E_1 G_1} du dv = \frac{du dv}{(1 - u^2)(1 - v^2)}.$$

For g_2 ,

$$E_2(u, v) = \frac{1}{(1 - v^2)^2}, \quad G_2(u, v) = \frac{1}{(1 - u^2)^2},$$

whence

$$dA_{g_2} = \sqrt{E_2 G_2} du dv = \frac{du dv}{(1 - u^2)(1 - v^2)}.$$

Thus $dA_{g_1} = dA_{g_2}$ pointwise on V , and the identity map $\text{id}_V : (V, g_1) \rightarrow (V, g_2)$ is a diffeomorphism preserving areas.

Properly embedded paths and finite length.

For g_2 consider the path

$$\gamma_2(t) = (\tanh t, 0), \quad t \in [0, \infty).$$

We have $|\tanh t| < 1$ and $\tanh t \rightarrow 1$ as $t \rightarrow \infty$, so γ_2 tends to the boundary point $(1, 0)$ and is properly embedded in V . Along γ_2 , $v \equiv 0$ and $du = \tanh'(t) dt = \text{sech}^2 t dt$, so for

$$g_2 = \frac{du^2}{(1 - v^2)^2} + \frac{dv^2}{(1 - u^2)^2}$$

we obtain $dv = 0$ and $1 - v^2 = 1$, hence $ds_{g_2} = |du|$ and

$$L_{g_2}(\gamma_2) = \int_0^\infty |\tanh'(t)| dt = \int_0^\infty \text{sech}^2 t dt = [\tanh t]_0^\infty = 1 < \infty.$$

Thus (V, g_2) admits a properly embedded path of finite length.

Now take any properly embedded path $\gamma_1(t) = (u(t), v(t)) : [0, \infty) \rightarrow V$ for g_1 . Because V is bounded and γ_1 is proper, its image must escape every compact subset, hence approach the boundary ∂V . Therefore at least one of the coordinates tends to ± 1 as $t \rightarrow \infty$; without loss of generality assume $u(t) \rightarrow 1$.

The speed in the metric g_1 is

$$|\gamma_1'(t)|_{g_1} = \sqrt{\frac{u'(t)^2}{(1 - u(t)^2)^2} + \frac{v'(t)^2}{(1 - v(t)^2)^2}} \geq \frac{|u'(t)|}{1 - u(t)^2}.$$

On any interval where u is monotone increasing from u_0 to u_1 we have

$$\int \frac{|u'(t)|}{1 - u(t)^2} dt = \int_{u_0}^{u_1} \frac{du}{1 - u^2}.$$

As $u_1 \rightarrow 1$, this integral diverges to $+\infty$, so by choosing t large we can force the partial length of γ_1 to be arbitrarily large. Hence

$$L_{g_1}(\gamma_1) = \int_0^\infty |\gamma_1'(t)|_{g_1} dt = \infty.$$

Thus every properly embedded path in (V, g_1) has infinite length.

Non-isometry. Suppose there were an isometry $\Phi : (V, g_1) \rightarrow (V, g_2)$. By part (i), Φ would carry properly embedded paths to properly embedded paths, and by definition of isometry it would preserve their lengths. But (V, g_2) admits a properly embedded path of finite length, while no such path exists in (V, g_1) . This is a contradiction. Therefore (V, g_1) and (V, g_2) are not isometric.

Visualizing the metrics of Problem 8

In this subsection we illustrate the abstract Riemannian metrics from Problem 8 by plotting their most important scale factors. This helps to see geometrically why distances can remain bounded while areas become unbounded, and how the two metrics on the square V are related by an isometry.

(a) The metric on the disc

On the open unit disc $B(0, 1) \subset \mathbb{R}^2$ we are given the metric

$$g_{\mathbb{D}} = \frac{du^2 + dv^2}{1 - u^2 - v^2}.$$

It is conformal to the Euclidean metric: the length element is

$$ds = \lambda(u, v) \sqrt{du^2 + dv^2}, \quad \lambda(u, v) = \frac{1}{\sqrt{1 - u^2 - v^2}},$$

and the area element is

$$dA = A(u, v) du dv, \quad A(u, v) = \frac{1}{1 - u^2 - v^2}.$$

The factor λ tells us how much *lengths* are stretched, while A tells us how much *areas* are stretched, compared to the standard Euclidean structure on the disc.

Notice that as $u^2 + v^2 \rightarrow 1$ (approaching the boundary circle), both λ and A blow up; however, their behaviour in integrals is different: the radial integral of λ from 0 to 1 converges (finite distances to the boundary), whereas the radial integral of A diverges (infinite area near the boundary).

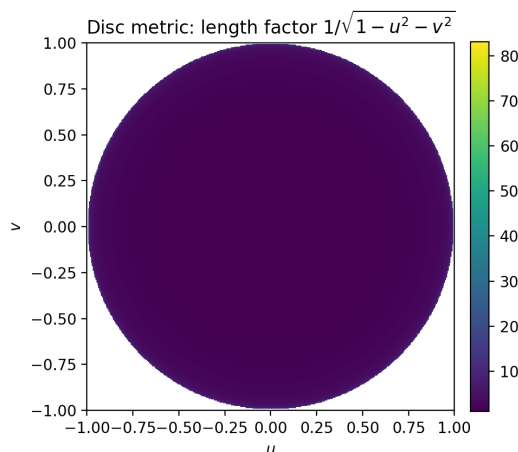


Figure 3: Unit disc with the conformal factor for lengths $\lambda(u, v) = 1/\sqrt{1 - u^2 - v^2}$. Colours indicate how distances are stretched; the factor blows up near the boundary $u^2 + v^2 = 1$, but geodesic diameters remain finite.

In the picture above, the colour intensity increases as one approaches the circle $u^2 + v^2 = 1$, reflecting that any small Euclidean segment near the boundary is scaled by a

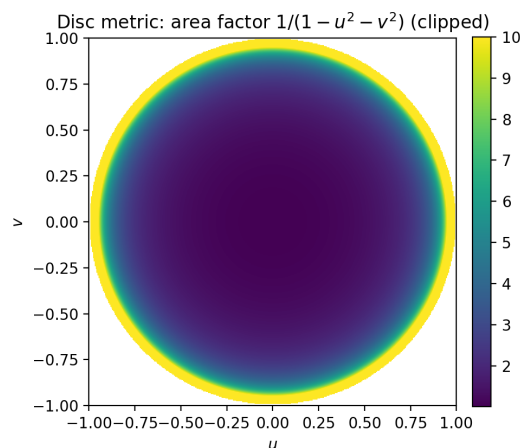


Figure 4: Unit disc with the area factor $A(u, v) = 1/(1 - u^2 - v^2)$ (clipped in the picture). This diverges near the boundary, illustrating why areas can be unbounded even though distances are bounded.

large factor in the Riemannian metric. Radial segments towards the boundary are still of finite total length because the integral of λ along a radius converges.

The second picture shows the area factor $A(u, v)$. The extremely bright ring near the boundary represents the region where the metric inflates area so strongly that the total area of the disc (in this metric) becomes infinite, even though the metric diameter of the disc is finite.

(b) The two metrics on the square V

Let $V = \{(u, v) \in \mathbb{R}^2 : |u| < 1, |v| < 1\}$ and define

$$g^{(1)} = \frac{du^2}{(1 - u^2)^2} + \frac{dv^2}{(1 - v^2)^2}, \quad g^{(2)} = \frac{du^2}{(1 - v^2)^2} + \frac{dv^2}{(1 - u^2)^2}.$$

Both metrics are *product-type*: the u - and v -directions stay orthogonal, but the scale factors in these directions depend separately on u and v . Each factor $(1 - u^2)^{-2}$ or $(1 - v^2)^{-2}$ blows up as one approaches the corresponding boundary line $u = \pm 1$ or $v = \pm 1$. Thus the boundary of the square is “pushed to infinity” in the associated metric.

For $g^{(1)}$ we have

$$g_{11}^{(1)} = \frac{1}{(1 - u^2)^2}, \quad g_{22}^{(1)} = \frac{1}{(1 - v^2)^2},$$

so the difference $g_{11}^{(1)} - g_{22}^{(1)}$ measures the anisotropy between the horizontal and vertical directions. Near $u = \pm 1$ the horizontal direction becomes extremely costly, while near $v = \pm 1$ it is the vertical direction that is strongly stretched.

For $g^{(2)}$ the roles of u and v are swapped:

$$g_{11}^{(2)} = \frac{1}{(1 - v^2)^2}, \quad g_{22}^{(2)} = \frac{1}{(1 - u^2)^2}.$$

Consequently, the anisotropy pattern $g_{11}^{(2)} - g_{22}^{(2)}$ is the mirror image of the previous one under the swap $(u, v) \mapsto (v, u)$.

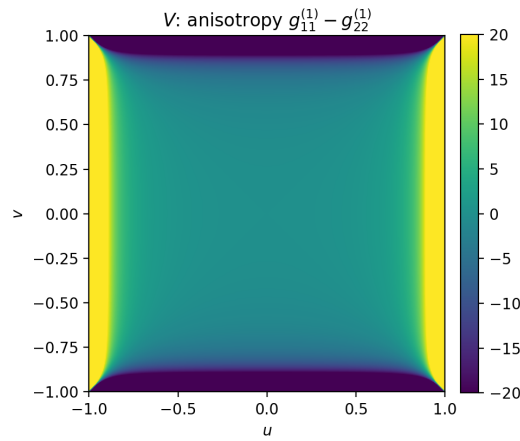


Figure 5: Open square $V = \{|u| < 1, |v| < 1\}$ with metric $g^{(1)} = \frac{du^2}{(1-u^2)^2} + \frac{dv^2}{(1-v^2)^2}$. The colours show the anisotropy $g_{11}^{(1)} - g_{22}^{(1)}$: near $u = \pm 1$ the u -direction is strongly stretched, near $v = \pm 1$ the v -direction dominates.

Indeed, the coordinate switch

$$\Phi : V \rightarrow V, \quad \Phi(u, v) = (v, u),$$

sends $g^{(1)}$ to $g^{(2)}$, so it defines an isometry between the two abstract Riemannian surfaces. The last two charts make this visually clear: the colour patterns are identical up to the interchange of the horizontal and vertical axes.

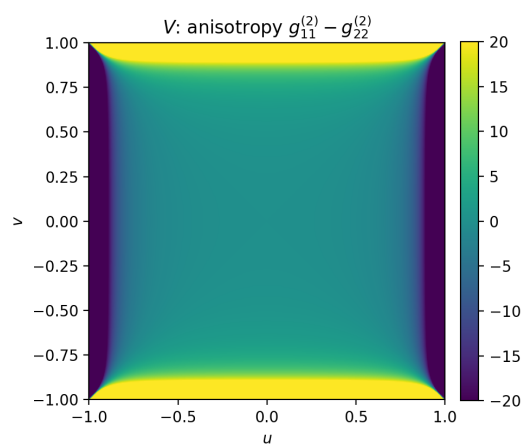


Figure 6: The same domain with metric $g^{(2)} = \frac{du^2}{(1-v^2)^2} + \frac{dv^2}{(1-u^2)^2}$. The anisotropy $g_{11}^{(2)} - g_{22}^{(2)}$ is essentially the $u \leftrightarrow v$ swap of the previous figure, reflecting that the map $(u, v) \mapsto (v, u)$ is an isometry between $g^{(1)}$ and $g^{(2)}$.

Examples for (ii): non-isometry and an area-preserving map

We now illustrate statement (ii) for the two Riemannian metrics on

$$V = \{(u, v) \in \mathbb{R}^2 : |u| < 1, |v| < 1\},$$

namely

$$g^{(1)} = \frac{du^2}{(1-u^2)^2} + \frac{dv^2}{(1-v^2)^2}, \quad g^{(2)} = \frac{du^2}{(1-v^2)^2} + \frac{dv^2}{(1-u^2)^2}.$$

The pictures above for $g^{(1)}$ and $g^{(2)}$ show how each metric stretches the horizontal and vertical directions differently near the boundary $|u| = 1$ and $|v| = 1$.

Properly embedded paths and non-isometry

Recall that a *properly embedded path* in V is a map

$$\gamma : [0, \infty) \rightarrow V$$

such that the preimage of every compact set $K \subset V$ is compact (i.e. $\gamma^{-1}(K)$ is closed and bounded in $[0, \infty)$). Since V is a bounded open subset of $\mathbb{R}^2$, this means that a properly embedded path must “run off” towards the boundary of V as $t \rightarrow \infty$; it cannot accumulate in the interior.

The first metric $g^{(1)}$: no finite-length proper rays. For $g^{(1)}$ we can separate the u and v directions by the change of variables

$$x = \operatorname{arctanh}(u), \quad y = \operatorname{arctanh}(v).$$

Then

$$dx = \frac{du}{1-u^2}, \quad dy = \frac{dv}{1-v^2},$$

so

$$g^{(1)} = \frac{du^2}{(1-u^2)^2} + \frac{dv^2}{(1-v^2)^2} = dx^2 + dy^2.$$

Thus $(V, g^{(1)})$ is isometric to the Euclidean plane $(\mathbb{R}^2, dx^2 + dy^2)$ via the diffeomorphism $\Phi(u, v) = (\operatorname{arctanh} u, \operatorname{arctanh} v)$.

In a complete Riemannian manifold such as $\mathbb{R}^2$ with the Euclidean metric, any curve of *finite* length has a limit point: if $\gamma : [0, \infty) \rightarrow \mathbb{R}^2$ has finite length L , then it is a Cauchy curve and converges to some point in $\mathbb{R}^2$. Its image is contained in a closed ball and hence in a compact set. The preimage of this compact set is all of $[0, \infty)$, which is not compact, so γ cannot be proper.

Therefore $(\mathbb{R}^2, dx^2 + dy^2)$ (and hence $(V, g^{(1)})$) does not contain any properly embedded path of finite length. This is consistent with the chart for $g^{(1)}$: both horizontal and vertical directions are stretched so strongly near $|u| = 1$ and $|v| = 1$ that the boundary lies at infinite distance.

The second metric $g^{(2)}$: a finite-length proper ray. In contrast, $(V, g^{(2)})$ does admit a properly embedded path of finite length. Consider the vertical path

$$\gamma(t) = (u(t), 0), \quad u(t) = 1 - e^{-t}, \quad t \in [0, \infty).$$

Then $\gamma(t) \in V$ for all t and $u(t) \nearrow 1$ as $t \rightarrow \infty$, so the path runs out towards the boundary line $u = 1$.

Along this path we have $v(t) \equiv 0$ and $v'(t) = 0$, so the metric $g^{(2)}$ restricts to

$$ds^2 = \frac{du^2}{(1 - v^2)^2} = \frac{du^2}{(1 - 0)^2} = du^2.$$

Since $u'(t) = e^{-t}$, the speed of γ is

$$\|\gamma'(t)\|_{g^{(2)}} = |u'(t)| = e^{-t},$$

and its total length is

$$L_{g^{(2)}}(\gamma) = \int_0^\infty e^{-t} dt = 1 < \infty.$$

To see that γ is properly embedded, let $K \subset V$ be any compact set. There exists $\delta > 0$ such that

$$K \subset \{(u, v) \in V : |u| \leq 1 - \delta, |v| \leq 1 - \delta\}.$$

Choose $T > 0$ with $u(T) > 1 - \delta$; then $u(t) > 1 - \delta$ for all $t \geq T$, so $\gamma(t) \notin K$ for $t \geq T$. Thus $\gamma^{-1}(K) \subset [0, T]$, which is compact. Hence γ is a properly embedded path of finite length.

By part (i), any homeomorphism $V \rightarrow V$ (in particular, any isometry) must map properly embedded paths to properly embedded paths. Since $(V, g^{(1)})$ has no finite-length proper path but $(V, g^{(2)})$ does, the two Riemannian surfaces cannot be isometric.

The charts for $g^{(2)}$ above illustrate this behaviour: along the vertical line $v = 0$ the coefficient in front of du^2 is $1/(1 - v^2)^2 = 1$, so the boundary $u = 1$ is at finite distance, whereas for $g^{(1)}$ the same vertical direction is heavily stretched near $u = \pm 1$, placing the boundary at infinite distance.

Area forms and an area-preserving diffeomorphism

Although the metrics are not isometric, they induce the *same* area form on V . For a diagonal metric

$$g = A(u, v) du^2 + B(u, v) dv^2,$$

the area element is

$$dA_g = \sqrt{\det g} du dv = \sqrt{A(u, v)B(u, v)} du dv.$$

For $g^{(1)}$ we have

$$A_1(u, v) = \frac{1}{(1 - u^2)^2}, \quad B_1(u, v) = \frac{1}{(1 - v^2)^2},$$

so

$$dA_{g^{(1)}} = \frac{1}{(1 - u^2)(1 - v^2)} du dv.$$

For $g^{(2)}$,

$$A_2(u, v) = \frac{1}{(1 - v^2)^2}, \quad B_2(u, v) = \frac{1}{(1 - u^2)^2},$$

whence

$$dA_{g^{(2)}} = \sqrt{A_2 B_2} du dv = \frac{1}{(1 - u^2)(1 - v^2)} du dv = dA_{g^{(1)}}.$$

Thus the identity map

$$\text{id} : (V, g^{(1)}) \longrightarrow (V, g^{(2)})$$

is an *area-preserving diffeomorphism*: it is a smooth bijection with smooth inverse, and it pulls back the area form of $g^{(2)}$ to that of $g^{(1)}$.

In summary, the two metrics on V have the same area density (as reflected in the similar colour patterns in the charts), but differ in their large-scale geometry: one is complete, the other admits a properly embedded path of finite length. This shows that area preservation is a much weaker condition than Riemannian isometry.

Diameters and bounded distances in the disc metric

On the open unit disc $B(0, 1) \subset \mathbb{R}^2$ we consider the abstract Riemannian metric

$$g = \frac{du^2 + dv^2}{1 - u^2 - v^2}.$$

Introducing polar coordinates $u = r \cos \theta$, $v = r \sin \theta$ with $0 \leq r < 1$, $0 \leq \theta < 2\pi$, we obtain

$$du^2 + dv^2 = dr^2 + r^2 d\theta^2,$$

so

$$ds^2 = \frac{dr^2 + r^2 d\theta^2}{1 - r^2}.$$

Diameters are length-minimizing. Let p and q be two points lying on the same diameter; in polar coordinates we may assume

$$p = (r_1, \theta_0), \quad q = (r_2, \theta_0), \quad 0 \leq r_1 < r_2 < 1.$$

Consider any smooth curve

$$\gamma(t) = (r(t), \theta(t)), \quad t \in [a, b],$$

with $\gamma(a) = p$, $\gamma(b) = q$. Its speed with respect to g is

$$\|\gamma'(t)\|_g = \sqrt{\frac{r'(t)^2 + r(t)^2 \theta'(t)^2}{1 - r(t)^2}} \geq \frac{|r'(t)|}{\sqrt{1 - r(t)^2}},$$

because $r(t)^2 \theta'(t)^2 \geq 0$.

Therefore the length of γ satisfies

$$L_g(\gamma) = \int_a^b \|\gamma'(t)\|_g dt \geq \int_a^b \frac{|r'(t)|}{\sqrt{1 - r(t)^2}} dt.$$

If γ is monotone in r (which we may assume after reparametrising), then r runs from r_1 to r_2 , and we obtain

$$L_g(\gamma) \geq \int_{r_1}^{r_2} \frac{dr}{\sqrt{1-r^2}}.$$

Now consider the *radial segment* along the diameter,

$$\gamma_0(t) = (r(t), \theta(t)) = (t, \theta_0), \quad t \in [r_1, r_2].$$

For this curve we have $\theta'(t) \equiv 0$, hence

$$\|\gamma'_0(t)\|_g = \sqrt{\frac{1}{1-t^2}} = \frac{1}{\sqrt{1-t^2}},$$

and its length is

$$L_g(\gamma_0) = \int_{r_1}^{r_2} \frac{dt}{\sqrt{1-t^2}} = \arcsin(r_2) - \arcsin(r_1).$$

We see that γ_0 attains equality in the inequality above, so *among all curves joining p and q the radial segment is of minimal length*. Thus diameters (radial segments at fixed angle) are length-minimizing geodesics.

Bounded distances. In particular, the distance from the centre to a boundary point is

$$d_g((0, 0), (1, \theta_0)) = \int_0^1 \frac{dr}{\sqrt{1-r^2}} = \left[\arcsin(r) \right]_0^1 = \frac{\pi}{2}.$$

So every point of the disc lies at distance at most $\pi/2$ from the origin.

For any two points $p, q \in B(0, 1)$, the triangle inequality gives

$$d_g(p, q) \leq d_g(p, 0) + d_g(0, q) \leq \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

Hence the metric diameter of $(B(0, 1), g)$ is at most π . In fact, two opposite boundary points on a diameter (for instance $(r = 1, \theta)$ and $(r = 1, \theta + \pi)$) are joined by a length-minimizing path consisting of two radial segments, and the total length of this diameter is

$$L_{\text{diameter}} = 2 \cdot d_g(0, \partial B(0, 1)) = 2 \cdot \frac{\pi}{2} = \pi.$$

Visual interpretation. The conformal factor

$$\lambda(r) = \frac{1}{\sqrt{1-r^2}}$$

blows up as $r \rightarrow 1$, so small Euclidean segments near the boundary are greatly stretched in the Riemannian metric. Nevertheless, the integral $\int_0^1 \lambda(r) dr$ converges, so the boundary circle is at finite distance. This is reflected in the chart of the disc metric (see Figure 7), where colours become more intense near the boundary but radial geodesics still have finite total length.

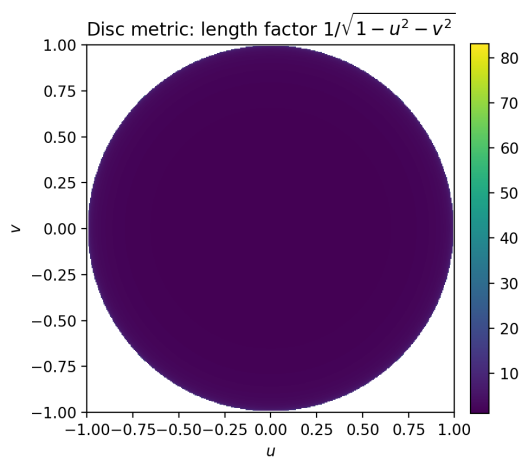


Figure 7: Unit disc with the conformal factor for lengths $\lambda(u, v) = 1/\sqrt{1-u^2-v^2}$. Colours indicate how distances are stretched; the factor blows up near the boundary $u^2 + v^2 = 1$, but radial diameters still have finite total length.

E1. Exponential cone and helicoid

Show that the surfaces Σ and Σ' in $\mathbb{R}^3$ defined as the images of

$$\sigma(u, v) = (u \cos v, u \sin v, \ln u), \quad \tau(u, v) = (u \cos v, u \sin v, v),$$

(where $u > 0$ and $v > 0$) have the same Gauss curvature, but are not locally isometric.

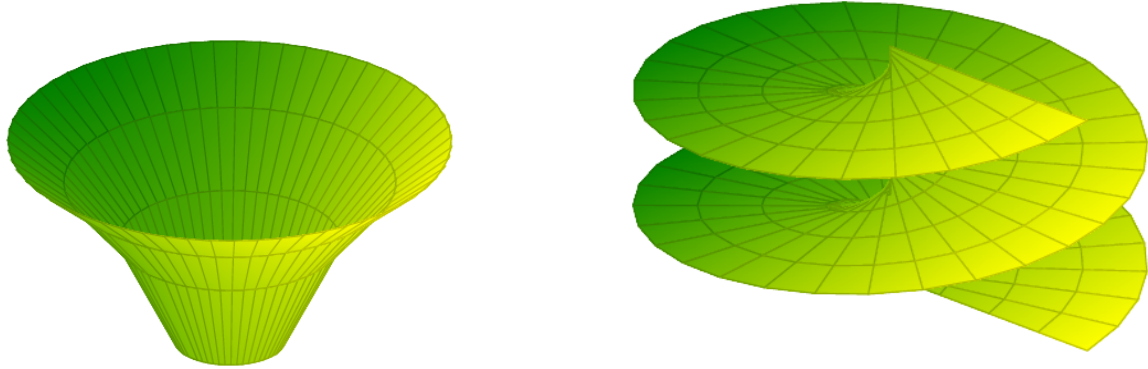


Figure 8: The surfaces Σ (left) and Σ' (right): an ‘exponential cone’ and a helicoid.

First fundamental forms. For the exponential cone Σ ,

$$\sigma(u, v) = (u \cos v, u \sin v, \ln u),$$

we have

$$\sigma_u = (\cos v, \sin v, 1/u), \quad \sigma_v = (-u \sin v, u \cos v, 0).$$

Thus

$$E_\Sigma = \langle \sigma_u, \sigma_u \rangle = 1 + \frac{1}{u^2}, \quad F_\Sigma = \langle \sigma_u, \sigma_v \rangle = 0, \quad G_\Sigma = \langle \sigma_v, \sigma_v \rangle = u^2,$$

and the first fundamental form is

$$I_\Sigma = \left(1 + \frac{1}{u^2}\right) du^2 + u^2 dv^2.$$

For the helicoid Σ' ,

$$\tau(u, v) = (u \cos v, u \sin v, v),$$

we obtain

$$\tau_u = (\cos v, \sin v, 0), \quad \tau_v = (-u \sin v, u \cos v, 1),$$

so

$$E_{\Sigma'} = 1, \quad F_{\Sigma'} = 0, \quad G_{\Sigma'} = u^2 + 1,$$

and

$$I_{\Sigma'} = du^2 + (u^2 + 1) dv^2.$$

Gauss curvature. For Σ we compute

$$\sigma_u \times \sigma_v = (-\cos v, -\sin v, u), \quad \|\sigma_u \times \sigma_v\|^2 = u^2 + 1,$$

so a unit normal is

$$n = \frac{1}{\sqrt{u^2 + 1}}(-\cos v, -\sin v, u).$$

Second derivatives are

$$\sigma_{uu} = (0, 0, -1/u^2), \quad \sigma_{uv} = (-\sin v, \cos v, 0), \quad \sigma_{vv} = (-u \cos v, -u \sin v, 0).$$

Hence

$$e_\Sigma = \langle \sigma_{uu}, n \rangle = -\frac{1}{u\sqrt{u^2 + 1}}, \quad f_\Sigma = \langle \sigma_{uv}, n \rangle = 0, \quad g_\Sigma = \langle \sigma_{vv}, n \rangle = \frac{u}{\sqrt{u^2 + 1}}.$$

With E_Σ, G_Σ as above and $F_\Sigma = 0$,

$$K_\Sigma = \frac{e_\Sigma g_\Sigma - f_\Sigma^2}{E_\Sigma G_\Sigma - F_\Sigma^2} = \frac{-\frac{1}{u\sqrt{u^2 + 1}} \cdot \frac{u}{\sqrt{u^2 + 1}}}{(1 + u^{-2})u^2} = -\frac{1}{(1 + u^2)^2}.$$

For Σ' we have

$$\tau_u \times \tau_v = (\sin v, -\cos v, u), \quad \|\tau_u \times \tau_v\|^2 = u^2 + 1,$$

so a unit normal is

$$n' = \frac{1}{\sqrt{u^2 + 1}}(\sin v, -\cos v, u).$$

Second derivatives:

$$\tau_{uu} = (0, 0, 0), \quad \tau_{uv} = (-\sin v, \cos v, 0), \quad \tau_{vv} = (-u \cos v, -u \sin v, 0).$$

Thus

$$e_{\Sigma'} = 0, \quad f_{\Sigma'} = \langle \tau_{uv}, n' \rangle = -\frac{1}{\sqrt{u^2 + 1}}, \quad g_{\Sigma'} = 0.$$

With $E_{\Sigma'} = 1, F_{\Sigma'} = 0, G_{\Sigma'} = u^2 + 1$,

$$K_{\Sigma'} = \frac{e_{\Sigma'} g_{\Sigma'} - f_{\Sigma'}^2}{E_{\Sigma'} G_{\Sigma'} - F_{\Sigma'}^2} = \frac{-\frac{1}{u^2 + 1}}{u^2 + 1} = -\frac{1}{(1 + u^2)^2}.$$

Hence

$$K_\Sigma(u, v) = K_{\Sigma'}(u, v) = -\frac{1}{(1 + u^2)^2},$$

so the two surfaces have the same Gauss curvature.

They are not locally isometric. Suppose there exists a local isometry $\Phi : \Sigma \supset U \rightarrow U' \subset \Sigma'$. Because an isometry preserves Gauss curvature and

$$K(u) = -\frac{1}{(1+u^2)^2}, \quad u > 0,$$

is strictly increasing ($K'(u) = 4u/(1+u^2)^3 > 0$), points with different u have different curvature. Thus Φ must preserve the coordinate u : if a point with coordinates (u, v) maps to $(\tilde{u}, \tilde{v})$, then $K(u) = K(\tilde{u})$ implies $\tilde{u} = u$.

Hence in (u, v) -coordinates we can write

$$\Phi(u, v) = (u, \varphi(u, v)).$$

Put $a = \varphi_u$ and $b = \varphi_v$. The Jacobian matrix is

$$D\Phi = \begin{pmatrix} 1 & 0 \\ a & b \end{pmatrix}.$$

The pullback of $I_{\Sigma'}$ under Φ has coefficients

$$(\Phi^* I_{\Sigma'})_{uu} = 1 + (u^2 + 1)a^2,$$

$$(\Phi^* I_{\Sigma'})_{uv} = (u^2 + 1)ab,$$

$$(\Phi^* I_{\Sigma'})_{vv} = (u^2 + 1)b^2.$$

These must equal $E_\Sigma, F_\Sigma, G_\Sigma$, i.e.

$$1 + \frac{1}{u^2} = 1 + (u^2 + 1)a^2, \quad 0 = (u^2 + 1)ab, \quad u^2 = (u^2 + 1)b^2.$$

From the middle equation, either $a = 0$ or $b = 0$. If $b = 0$ then $u^2 = 0$, impossible for $u > 0$. Thus $a = 0$, and the first equation gives $1 + 1/u^2 = 1$, also impossible. Therefore such a local isometry Φ cannot exist.

Consequently, the exponential cone Σ and the helicoid Σ' have the same Gauss curvature but are not locally isometric.

E2. Moving frame and Gauss' theorema egregium

Consider a smooth surface $\Sigma \subset \mathbb{R}^3$ with Gauss curvature κ , and an allowable parametrization

$$\sigma : V \rightarrow U \subset \Sigma$$

with first fundamental form

$$du^2 + G(u, v) dv^2.$$

Let

$$e = \sigma_u, \quad f = \frac{\sigma_v}{\sqrt{G}}, \quad n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|},$$

so that $\langle e, f, n \rangle$ is an orthonormal basis of $\mathbb{R}^3$ depending on $(u, v) \in V$ (a moving frame).

1. Show that $n = e \times f$.

Since

$$\sigma_u \cdot \sigma_u = 1, \quad \sigma_u \cdot \sigma_v = 0, \quad \sigma_v \cdot \sigma_v = G,$$

we have $|e| = 1$, $|f| = 1$, and $e \cdot f = 0$. Moreover,

$$\sigma_u \times \sigma_v = e \times (\sqrt{G} f) = \sqrt{G} (e \times f),$$

and $\|\sigma_u \times \sigma_v\| = \sqrt{G}$, so

$$n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = e \times f.$$

2. Frame derivative formulas.

The orthonormality relations

$$e \cdot e = f \cdot f = n \cdot n = 1, \quad e \cdot f = e \cdot n = f \cdot n = 0$$

imply, upon differentiating with respect to u and v , that e_u, f_u, e_v, f_v have no components along themselves. Thus there exist functions $\alpha, \beta, \lambda_i, \mu_j$ such that

$$e_u = \alpha f + \lambda_1 n, \quad e_v = \beta f + \mu_1 n, \quad f_u = -\alpha e + \lambda_2 n, \quad f_v = -\beta e + \mu_2 n.$$

3. Show $\alpha = 0$ and $\beta = (\sqrt{G})_u$.

Since $e = \sigma_u$ and $f = \sigma_v/\sqrt{G}$,

$$\alpha = e_u \cdot f = \sigma_{uu} \cdot \frac{\sigma_v}{\sqrt{G}} = \frac{1}{\sqrt{G}} (\sigma_{uu} \cdot \sigma_v).$$

Differentiating $\sigma_u \cdot \sigma_v = 0$ with respect to u gives $\sigma_{uu} \cdot \sigma_v + \sigma_u \cdot \sigma_{uv} = 0$. From $(\sigma_u \cdot \sigma_u)_v = 0$ we get $\sigma_{uv} \cdot \sigma_u = 0$, hence $\sigma_{uu} \cdot \sigma_v = 0$ and therefore $\alpha = 0$.

Similarly,

$$\beta = e_v \cdot f = \sigma_{uv} \cdot \frac{\sigma_v}{\sqrt{G}} = \frac{1}{\sqrt{G}} (\sigma_{uv} \cdot \sigma_v).$$

Differentiating $\sigma_v \cdot \sigma_v = G$ with respect to u yields

$$G_u = 2 \sigma_{uv} \cdot \sigma_v,$$

so

$$\beta = \frac{G_u}{2\sqrt{G}} = (\sqrt{G})_u.$$

4. Show $\lambda_1 \mu_2 - \lambda_2 \mu_1 = e_u \cdot f_v - f_u \cdot e_v = -\beta_u = -(\sqrt{G})_{uu}$.

Using the frame expansions and orthonormality,

$$e_u \cdot f_v = (\alpha f + \lambda_1 n) \cdot (-\beta e + \mu_2 n) = \lambda_1 \mu_2,$$

$$f_u \cdot e_v = (-\alpha e + \lambda_2 n) \cdot (\beta f + \mu_1 n) = \lambda_2 \mu_1.$$

Hence

$$e_u \cdot f_v - f_u \cdot e_v = \lambda_1 \mu_2 - \lambda_2 \mu_1.$$

Since $\beta = e_v \cdot f$, we have

$$\beta_u = (e_v \cdot f)_u = e_{vu} \cdot f + e_v \cdot f_u.$$

Using the equality $e_{uv} = e_{vu}$ together with the frame formulas and collecting coefficients in the orthonormal basis yields

$$\beta_u = \lambda_2 \mu_1 - \lambda_1 \mu_2 = -(\lambda_1 \mu_2 - \lambda_2 \mu_1).$$

Thus

$$\lambda_1 \mu_2 - \lambda_2 \mu_1 = e_u \cdot f_v - f_u \cdot e_v = -\beta_u.$$

Since $\beta = (\sqrt{G})_u$, it follows that

$$-\beta_u = -(\sqrt{G})_{uu}.$$

5. Use the Gauss map.

The Gauss map $n : \Sigma \rightarrow S^2$ satisfies

$$Dn(\sigma_u) = n_u, \quad Dn(\sigma_v) = n_v.$$

The restriction of Dn to the tangent plane is the shape operator whose determinant equals the Gauss curvature κ . In the oriented basis $\{e, f\}$ this gives

$$n_u \times n_v = \kappa (\sigma_u \times \sigma_v).$$

Taking the scalar product with n and using $n = e \times f$ we obtain

$$(n_u \times n_v) \cdot n = \kappa (\sigma_u \times \sigma_v) \cdot n = \kappa \|\sigma_u \times \sigma_v\| = \kappa \sqrt{G}.$$

On the other hand, expressing n_u and n_v in the moving frame and computing $(n_u \times n_v) \cdot n$ gives

$$(n_u \times n_v) \cdot n = \lambda_1 \mu_2 - \lambda_2 \mu_1 = -(\sqrt{G})_{uu}.$$

Therefore

$$\boxed{\kappa \sqrt{G} = -(\sqrt{G})_{uu}} \quad \Longrightarrow \quad \boxed{\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}}}.$$

Gauss' theorema egregium. The formula

$$\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}}$$

expresses the Gauss curvature solely in terms of the first fundamental form $du^2 + G(u, v) dv^2$. Hence κ depends only on the intrinsic metric of the surface. In particular, if two smooth embedded surfaces in $\mathbb{R}^3$ are isometric, their first fundamental forms agree, and therefore their Gauss curvatures coincide. This is Gauss' *theorema egregium*.

Examples of isometric and non-isometric surfaces with first fundamental form $du^2 + G(u, v) dv^2$

In this section we collect concrete examples illustrating the abstract setting of Problem E2, where the first fundamental form of a parametrized surface has the special form

$$I = du^2 + G(u, v) dv^2,$$

i.e. $E = 1$, $F = 0$ and $G = G(u, v)$. In all examples below we work with coordinates (u, v) on a parameter domain $V \subset \mathbb{R}^2$ and a parametrization $\sigma : V \rightarrow \Sigma \subset \mathbb{R}^3$.

Recall from E2 that for such an *allowable* parametrization the Gauss curvature satisfies

$$\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}}.$$

1. Isometric pair: plane strip and cylinder strip

We first consider a pair of surfaces which are locally isometric and whose first fundamental forms are both of the special form $du^2 + G dv^2$ with the same function G .

Plane strip. Let

$$\sigma_{\text{pl}}(u, v) = (u, v, 0), \quad (u, v) \in (0, 2\pi) \times (-L, L)$$

for some fixed $L > 0$. Then

$$\sigma_{\text{pl},u} = (1, 0, 0), \quad \sigma_{\text{pl},v} = (0, 1, 0),$$

so

$$E = \sigma_{\text{pl},u} \cdot \sigma_{\text{pl},u} = 1, \quad F = \sigma_{\text{pl},u} \cdot \sigma_{\text{pl},v} = 0, \quad G = \sigma_{\text{pl},v} \cdot \sigma_{\text{pl},v} = 1.$$

Hence the first fundamental form on this strip of the plane is

$$I_{\text{pl}} = du^2 + dv^2,$$

which is of the desired type with $G(u, v) \equiv 1$.

From Gauss' formula we obtain

$$\sqrt{G} = 1, \quad (\sqrt{G})_{uu} = 0 \quad \implies \quad \kappa_{\text{pl}} = 0.$$

Cylinder strip. Consider the unit cylinder and the parametrization

$$\sigma_{\text{cyl}}(u, v) = (\cos u, \sin u, v), \quad (u, v) \in (0, 2\pi) \times (-L, L).$$

Then

$$\sigma_{\text{cyl},u} = (-\sin u, \cos u, 0), \quad \sigma_{\text{cyl},v} = (0, 0, 1),$$

so

$$E = \sigma_{\text{cyl},u} \cdot \sigma_{\text{cyl},u} = 1, \quad F = \sigma_{\text{cyl},u} \cdot \sigma_{\text{cyl},v} = 0, \quad G = \sigma_{\text{cyl},v} \cdot \sigma_{\text{cyl},v} = 1.$$

Hence

$$I_{\text{cyl}} = du^2 + dv^2,$$

again with $G(u, v) \equiv 1$. The Gauss curvature is therefore $\kappa_{\text{cyl}} = 0$ by the same computation as above.

Isometry between the strip and the cylinder. On the common parameter domain $V = (0, 2\pi) \times (-L, L)$, the identity map

$$\Phi : V \rightarrow V, \quad \Phi(u, v) = (u, v),$$

satisfies

$$\Phi^* I_{\text{cyl}} = I_{\text{pl}},$$

because their metric coefficients E, F, G coincide identically. Thus σ_{pl} and σ_{cyl} describe locally isometric surfaces: a rectangular strip of the plane and a corresponding strip of the cylinder. Geometrically, one may “roll” the strip into the cylinder without stretching.

2. Non-isometric pair: plane (polar form) vs unit sphere

We now consider two surfaces whose first fundamental forms are both of the form $du^2 + G(u, v) dv^2$, but with different G and therefore different Gauss curvature. By the theorem egregium they cannot be locally isometric.

Plane in polar coordinates. On the plane (minus the origin) we use polar coordinates:

$$\sigma_{\mathbb{R}^2}(u, v) = (u \cos v, u \sin v, 0), \quad u > 0, v \in (0, 2\pi).$$

Then

$$\sigma_u = (\cos v, \sin v, 0), \quad \sigma_v = (-u \sin v, u \cos v, 0),$$

so

$$E = \sigma_u \cdot \sigma_u = 1, \quad F = \sigma_u \cdot \sigma_v = 0, \quad G = \sigma_v \cdot \sigma_v = u^2.$$

Hence

$$I_{\mathbb{R}^2} = du^2 + u^2 dv^2.$$

We compute the Gauss curvature using the E2 formula:

$$\sqrt{G} = u, \quad (\sqrt{G})_u = 1, \quad (\sqrt{G})_{uu} = 0,$$

and therefore

$$\kappa_{\mathbb{R}^2} = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}} = 0,$$

as expected for the plane.

Unit sphere. On the unit sphere $S^2 \subset \mathbb{R}^3$ take spherical coordinates:

$$\sigma_{S^2}(u, v) = (\sin u \cos v, \sin u \sin v, \cos u), \quad u \in (0, \pi), v \in (0, 2\pi).$$

We compute

$$\sigma_u = (\cos u \cos v, \cos u \sin v, -\sin u), \quad \sigma_v = (-\sin u \sin v, \sin u \cos v, 0),$$

and thus

$$E = \sigma_u \cdot \sigma_u = 1, \quad F = \sigma_u \cdot \sigma_v = 0, \quad G = \sigma_v \cdot \sigma_v = \sin^2 u.$$

The first fundamental form is

$$I_{S^2} = du^2 + \sin^2 u \, dv^2.$$

Again applying the curvature formula,

$$\sqrt{G} = \sin u, \quad (\sqrt{G})_u = \cos u, \quad (\sqrt{G})_{uu} = -\sin u,$$

we obtain

$$\kappa_{S^2} = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}} = -\frac{-\sin u}{\sin u} = 1.$$

Non-isometry. We have shown that

$$\kappa_{\mathbb{R}^2} = 0, \quad \kappa_{S^2} = 1.$$

Since Gauss curvature is an intrinsic invariant of the metric, there can be no local isometry between any patch of the plane (with the Euclidean metric) and any patch of the unit sphere (with the round metric).

3. Same curvature but not locally isometric: exponential cone vs helicoid (from E1)

For completeness we recall the example from Problem E1: two surfaces with the same Gauss curvature, but different first fundamental forms, hence not locally isometric.

Exponential cone. Let

$$\sigma_{\Sigma}(u, v) = (u \cos v, u \sin v, \ln u), \quad u > 0, v > 0.$$

One computes

$$I_{\Sigma} = \left(1 + \frac{1}{u^2}\right) du^2 + u^2 dv^2,$$

and

$$K_{\Sigma}(u, v) = -\frac{1}{(1 + u^2)^2}.$$

Helicoid. Let

$$\sigma_{\Sigma'}(u, v) = (u \cos v, u \sin v, v), \quad u > 0, v > 0.$$

Then

$$I_{\Sigma'} = du^2 + (u^2 + 1) dv^2,$$

and again

$$K_{\Sigma'}(u, v) = -\frac{1}{(1 + u^2)^2}.$$

Conclusion. Although Σ and Σ' have the same Gauss curvature function, their first fundamental forms differ:

$$I_{\Sigma} = \left(1 + \frac{1}{u^2}\right) du^2 + u^2 dv^2 \neq du^2 + (u^2 + 1) dv^2 = I_{\Sigma'}.$$

A direct calculation (cf. E1) shows that there is no local isometry between them. Thus equal Gauss curvature, by itself, does not guarantee local isometry; one needs the full metric.

We work with an allowable parametrization

$$\sigma : V \rightarrow \Sigma \subset \mathbb{R}^3, \quad I = du^2 + G(u, v) dv^2,$$

so that

$$\sigma_u \cdot \sigma_u = 1, \quad \sigma_u \cdot \sigma_v = 0, \quad \sigma_v \cdot \sigma_v = G(u, v) > 0.$$

Define

$$e = \sigma_u, \quad f = \frac{\sigma_v}{\sqrt{G}}, \quad n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}.$$

Then

$$\begin{aligned} \|e\|^2 &= e \cdot e = \sigma_u \cdot \sigma_u = 1, \\ \|f\|^2 &= f \cdot f = \frac{\sigma_v \cdot \sigma_v}{G} = \frac{G}{G} = 1, \end{aligned}$$

and

$$e \cdot f = \sigma_u \cdot \frac{\sigma_v}{\sqrt{G}} = \frac{\sigma_u \cdot \sigma_v}{\sqrt{G}} = \frac{0}{\sqrt{G}} = 0.$$

Thus e and f form an orthonormal basis of the tangent plane $T_{\sigma(u,v)}\Sigma$.

Moreover,

$$\sigma_u \times \sigma_v = e \times (\sqrt{G} f) = \sqrt{G} (e \times f),$$

so

$$\|\sigma_u \times \sigma_v\| = \sqrt{G} \|e \times f\| = \sqrt{G},$$

and hence

$$n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = \frac{\sqrt{G} (e \times f)}{\sqrt{G}} = e \times f.$$

Therefore (e, f, n) is an orthonormal frame in $\mathbb{R}^3$ along the surface, consisting of two orthonormal tangent vectors e, f and the unit normal n .

In these coordinates the first fundamental form is

$$I = du^2 + G(u, v) dv^2,$$

which means that the coordinate curves $u \mapsto \sigma(u, v_0)$ are parametrized by arc length ($E = 1$), the u - and v -curves are orthogonal ($F = 0$), and the scale factor in the v -direction is $\sqrt{G(u, v)}$.

Example 1.9 (Sphere). On the unit sphere with parametrization

$$\sigma(u, v) = (\sin u \cos v, \sin u \sin v, \cos u),$$

one finds

$$I = du^2 + \sin^2 u dv^2,$$

so $G(u, v) = \sin^2 u$. The corresponding moving frame $e = \sigma_u$, $f = \sigma_v/\sqrt{G}$, $n = e \times f$ is adapted to the meridians and parallels.

Example 1.10 (Plane in polar coordinates). On the plane (minus the origin) with

$$\sigma(u, v) = (u \cos v, u \sin v, 0),$$

we obtain

$$I = du^2 + u^2 dv^2,$$

so $G(u, v) = u^2$, and again $e = \sigma_u$, $f = \sigma_v/\sqrt{G}$, $n = e \times f$ provide an orthonormal frame adapted to radial lines and circles.

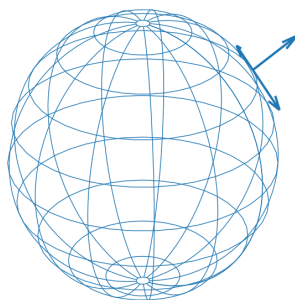


Figure 9: An allowable chart on the unit sphere. The vectors $e = \sigma_u$, $f = \sigma_v/\sqrt{G}$ and $n = e \times f$ form an orthonormal moving frame along the surface.

Stereographic chart and moving frame

Stereographic projection from the north pole gives a parametrization

$$\sigma : \mathbb{R}^2 \longrightarrow S^2 \setminus \{(0, 0, 1)\}$$

defined by

$$\sigma(x, y) = \left(\frac{2x}{x^2 + y^2 + 1}, \frac{2y}{x^2 + y^2 + 1}, \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1} \right).$$

Compute the partial derivatives:

$$\sigma_x = \frac{\partial \sigma}{\partial x}, \quad \sigma_y = \frac{\partial \sigma}{\partial y}.$$

One can check that the first fundamental form is

$$E = \sigma_x \cdot \sigma_x = \frac{4}{(1 + x^2 + y^2)^2}, \quad F = \sigma_x \cdot \sigma_y = 0,$$

$$G = \sigma_y \cdot \sigma_y = \frac{4}{(1 + x^2 + y^2)^2},$$

so

$$I = \frac{4}{(1 + x^2 + y^2)^2} (dx^2 + dy^2).$$

This is *not* of the “E2 special form” $du^2 + G(u, v) dv^2$ (here $E = G \neq 1$), but we can still build an orthonormal frame by normalizing the coordinate vectors:

$$e = \frac{\sigma_x}{\|\sigma_x\|}, \quad f = \frac{\sigma_y}{\|\sigma_y\|}, \quad n = e \times f.$$

Then (e, f, n) is again an orthonormal moving frame along the surface, just with a different underlying chart.

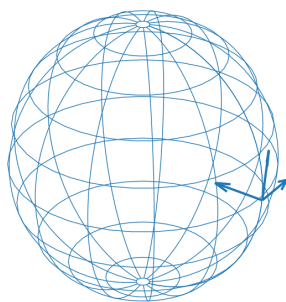


Figure 10: Unit sphere with a moving orthonormal frame (e, f, n) , constructed from stereographic coordinates (x, y) rather than spherical (polar) coordinates.

Chain rule interpretation of $Dn(\sigma_u) = n_u$. Let $\sigma : V \rightarrow \Sigma \subset \mathbb{R}^3$ be a parametrization and $N : \Sigma \rightarrow S^2$ the Gauss map. Define

$$n(u, v) := N(\sigma(u, v)).$$

Then by the chain rule,

$$n_u(u, v) = \frac{\partial}{\partial u} (N(\sigma(u, v))) = DN_{\sigma(u, v)}(\sigma_u(u, v)),$$

and similarly

$$n_v(u, v) = DN_{\sigma(u, v)}(\sigma_v(u, v)).$$

This is what is meant by

$$Dn(\sigma_u) = n_u, \quad Dn(\sigma_v) = n_v,$$

where Dn denotes the differential of the Gauss map restricted to the tangent plane.

Unit sphere in polar coordinates. For the unit sphere S^2 , the Gauss map is the identity $N(p) = p$. In spherical coordinates

$$\sigma(u, v) = (\sin u \cos v, \sin u \sin v, \cos u),$$

we have

$$n(u, v) = N(\sigma(u, v)) = \sigma(u, v),$$

hence

$$n_u = \sigma_u, \quad n_v = \sigma_v.$$

Since $DN_p = \text{Id}$, we obtain

$$DN_{\sigma(u,v)}(\sigma_u) = \sigma_u = n_u, \quad DN_{\sigma(u,v)}(\sigma_v) = \sigma_v = n_v.$$

Unit sphere in stereographic coordinates. Using stereographic coordinates

$$\sigma(x, y) = \left(\frac{2x}{x^2 + y^2 + 1}, \frac{2y}{x^2 + y^2 + 1}, \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1} \right),$$

we again have $N(p) = p$ and therefore $n(x, y) = \sigma(x, y)$, so

$$n_x = \sigma_x, \quad n_y = \sigma_y,$$

and

$$DN_{\sigma(x,y)}(\sigma_x) = \sigma_x = n_x, \quad DN_{\sigma(x,y)}(\sigma_y) = \sigma_y = n_y.$$

In both charts the equalities $Dn(\sigma_u) = n_u$, $Dn(\sigma_v) = n_v$ are simply the chain rule applied to $n = N \circ \sigma$.

Example: explicit matrix of Dn on the unit sphere

Consider the unit sphere $S^2 \subset \mathbb{R}^3$ with the usual spherical coordinates

$$\sigma(u, v) = (\sin u \cos v, \sin u \sin v, \cos u), \quad 0 < u < \pi, \quad 0 < v < 2\pi.$$

The Gauss map of S^2 is simply the identity

$$N(p) = p, \quad p \in S^2,$$

so in these coordinates

$$n(u, v) := N(\sigma(u, v)) = \sigma(u, v).$$

Tangent vectors and metric. We have

$$\sigma_u = (\cos u \cos v, \cos u \sin v, -\sin u), \quad \sigma_v = (-\sin u \sin v, \sin u \cos v, 0),$$

so

$$E = \sigma_u \cdot \sigma_u = 1, \quad F = \sigma_u \cdot \sigma_v = 0, \quad G = \sigma_v \cdot \sigma_v = \sin^2 u.$$

Thus the first fundamental form is

$$I = du^2 + \sin^2 u dv^2,$$

and an orthonormal tangent frame is given by

$$e := \sigma_u, \quad f := \frac{\sigma_v}{\sin u} \quad (u \in (0, \pi)).$$

Differential of the Gauss map. Since $n = \sigma$, we have

$$n_u = \sigma_u = e, \quad n_v = \sigma_v.$$

The differential Dn at the point $\sigma(u, v)$ is characterized by

$$Dn_{\sigma(u,v)}(\sigma_u) = n_u, \quad Dn_{\sigma(u,v)}(\sigma_v) = n_v.$$

We now express Dn in the orthonormal basis $\{e, f\}$ of the tangent plane. First,

$$Dn(e) = Dn(\sigma_u) = n_u = e.$$

Next, noting that $f = \sigma_v / \sin u$ and u is constant when we vary v ,

$$Dn(f) = Dn\left(\frac{1}{\sin u} \sigma_v\right) = \frac{1}{\sin u} Dn(\sigma_v) = \frac{1}{\sin u} n_v = \frac{1}{\sin u} \sigma_v = f.$$

Therefore, relative to the orthonormal basis $\{e, f\}$,

$$Dn|_{T_{\sigma(u,v)}S^2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Shape operator and Gauss curvature. By definition the shape operator is

$$S = -Dn|_{T\Sigma},$$

so in the basis $\{e, f\}$ we have

$$S = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its eigenvalues are both -1 , and hence the Gauss curvature is

$$K = \det S = (-1) \cdot (-1) = 1.$$

This agrees with the intrinsic formula for an allowable parametrization from Problem E2: here

$$G(u, v) = \sin^2 u, \quad \sqrt{G} = \sin u, \quad (\sqrt{G})_{uu} = -\sin u,$$

so

$$\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}} = -\frac{-\sin u}{\sin u} = 1.$$

Thus the determinant of the shape operator $S = -Dn$ coincides with the intrinsic Gauss curvature.

Example: Dn in stereographic coordinates on S^2

Consider the stereographic parametrization of the unit sphere (from the north pole) given by

$$\sigma : \mathbb{R}^2 \longrightarrow S^2 \setminus \{(0, 0, 1)\}, \quad \sigma(x, y) = \left(\frac{2x}{x^2 + y^2 + 1}, \frac{2y}{x^2 + y^2 + 1}, \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1} \right).$$

Set $r^2 = x^2 + y^2$ and $d = r^2 + 1$ for brevity.

First fundamental form. Compute the partial derivatives

$$\sigma_x = \frac{\partial \sigma}{\partial x}, \quad \sigma_y = \frac{\partial \sigma}{\partial y}.$$

A straightforward calculation shows

$$\sigma_x \cdot \sigma_x = \frac{4}{(1+x^2+y^2)^2}, \quad \sigma_y \cdot \sigma_y = \frac{4}{(1+x^2+y^2)^2}, \quad \sigma_x \cdot \sigma_y = 0.$$

Thus

$$E = \sigma_x \cdot \sigma_x = \frac{4}{(1+r^2)^2}, \quad F = 0, \quad G = \sigma_y \cdot \sigma_y = \frac{4}{(1+r^2)^2},$$

and the first fundamental form is

$$I = \frac{4}{(1+x^2+y^2)^2} (dx^2 + dy^2).$$

In particular, this chart is conformal: the metric is a scalar multiple of the Euclidean metric on $\mathbb{R}^2$.

Orthonormal tangent frame. Since $E = G$ and $F = 0$, the coordinate vectors σ_x and σ_y are orthogonal and have equal length

$$\|\sigma_x\|^2 = \|\sigma_y\|^2 = \frac{4}{(1+r^2)^2}.$$

We obtain an orthonormal tangent frame by normalizing:

$$e := \frac{\sigma_x}{\|\sigma_x\|}, \quad f := \frac{\sigma_y}{\|\sigma_y\|}, \quad n := e \times f.$$

Then $\{e, f\}$ is an orthonormal basis of $T_{\sigma(x,y)}S^2$ and n is the outward unit normal.

Gauss map and its differential. On the unit sphere, the Gauss map is simply

$$N(p) = p, \quad p \in S^2.$$

Therefore in these coordinates

$$n(x, y) := N(\sigma(x, y)) = \sigma(x, y),$$

hence

$$n_x = \sigma_x, \quad n_y = \sigma_y.$$

The differential of the Gauss map at the point $\sigma(x, y)$ is

$$Dn_{\sigma(x,y)} : T_{\sigma(x,y)}S^2 \longrightarrow T_{\sigma(x,y)}S^2.$$

Since N is the identity map on S^2 , its differential is the identity on each tangent space:

$$Dn_{\sigma(x,y)} = \text{Id}_{T_{\sigma(x,y)}S^2}.$$

In particular,

$$Dn(e) = e, \quad Dn(f) = f.$$

Thus, relative to the orthonormal basis $\{e, f\}$, the matrix of Dn is

$$Dn|_{T_{\sigma(x,y)}S^2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Shape operator and Gauss curvature. The shape operator is defined by

$$S = -Dn|_{TS^2}.$$

Hence, in the orthonormal frame $\{e, f\}$,

$$S = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its eigenvalues are both -1 , so the Gauss curvature is

$$K = \det S = (-1) \cdot (-1) = 1.$$

This agrees with the fact that the unit sphere has constant Gauss curvature $K \equiv 1$. We see explicitly that, even in stereographic coordinates where the metric is

$$I = \frac{4}{(1+x^2+y^2)^2} (dx^2 + dy^2),$$

the differential of the Gauss map restricted to the tangent plane has matrix $-I_2$ (up to sign) in an orthonormal frame, and its determinant gives the curvature.

Comparison of two charts on S^2 and the differential of the Gauss map

We compare two different parametrizations of the unit sphere $S^2 \subset \mathbb{R}^3$: spherical (polar) coordinates and stereographic coordinates. In both cases we compute the first fundamental form, an orthonormal frame, the differential of the Gauss map, and the Gauss curvature.

1. Spherical (polar) coordinates

Consider

$$\sigma(u, v) = (\sin u \cos v, \sin u \sin v, \cos u), \quad 0 < u < \pi, \quad 0 < v < 2\pi.$$

We have

$$\sigma_u = (\cos u \cos v, \cos u \sin v, -\sin u), \quad \sigma_v = (-\sin u \sin v, \sin u \cos v, 0),$$

so

$$E = \sigma_u \cdot \sigma_u = 1, \quad F = \sigma_u \cdot \sigma_v = 0, \quad G = \sigma_v \cdot \sigma_v = \sin^2 u,$$

and the first fundamental form is

$$I_{\text{pol}} = du^2 + \sin^2 u dv^2.$$

An orthonormal tangent frame is

$$e := \sigma_u, \quad f := \frac{\sigma_v}{\sin u}, \quad n := e \times f.$$

On the unit sphere the Gauss map is the identity, $N(p) = p$, hence $n(u, v) := N(\sigma(u, v)) = \sigma(u, v)$ and

$$n_u = \sigma_u, \quad n_v = \sigma_v.$$

The differential $Dn_{\sigma(u,v)}$ restricted to the tangent plane satisfies

$$Dn(e) = e, \quad Dn(f) = f,$$

so in the orthonormal basis $\{e, f\}$,

$$Dn|_{T_{\sigma(u,v)}S^2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The shape operator is $S = -Dn|_{TS^2}$, hence

$$S = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad K = \det S = 1.$$

2. Stereographic coordinates

Now consider stereographic projection from the north pole. With $r^2 = x^2 + y^2$ we define

$$\sigma(x, y) = \left(\frac{2x}{r^2 + 1}, \frac{2y}{r^2 + 1}, \frac{r^2 - 1}{r^2 + 1} \right), \quad (x, y) \in \mathbb{R}^2.$$

A direct computation shows

$$\sigma_x \cdot \sigma_x = \frac{4}{(1 + r^2)^2}, \quad \sigma_y \cdot \sigma_y = \frac{4}{(1 + r^2)^2}, \quad \sigma_x \cdot \sigma_y = 0.$$

Thus

$$E = G = \frac{4}{(1 + r^2)^2}, \quad F = 0,$$

and

$$I_{\text{stereo}} = \frac{4}{(1 + x^2 + y^2)^2} (dx^2 + dy^2),$$

so this chart is conformal.

We obtain an orthonormal tangent frame by normalizing the coordinate vectors:

$$e := \frac{\sigma_x}{\|\sigma_x\|}, \quad f := \frac{\sigma_y}{\|\sigma_y\|}, \quad n := e \times f.$$

Again, the Gauss map is $N(p) = p$, so

$$n(x, y) := N(\sigma(x, y)) = \sigma(x, y), \quad n_x = \sigma_x, \quad n_y = \sigma_y.$$

Since N is the identity on S^2 , its differential is the identity on each tangent space:

$$Dn_{\sigma(x,y)} = \text{Id}_{T_{\sigma(x,y)}S^2}.$$

In particular,

$$Dn(e) = e, \quad Dn(f) = f,$$

so in the orthonormal basis $\{e, f\}$ we again have

$$Dn|_{T_{\sigma(x,y)}S^2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S = -Dn = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad K = \det S = 1.$$

3. Comparison

In spherical coordinates the metric is

$$I_{\text{pol}} = du^2 + \sin^2 u \, dv^2,$$

while in stereographic coordinates it is

$$I_{\text{stereo}} = \frac{4}{(1 + x^2 + y^2)^2} (dx^2 + dy^2).$$

These two expressions look very different: one is adapted to meridians and parallels, the other is conformal to the Euclidean plane. Nevertheless, in each chart we can construct an orthonormal frame (e, f) and compute the matrix of the differential of the Gauss map $Dn|_{TS^2}$. In both cases this matrix is $-I_2$ for the shape operator, hence

$$K = \det S = 1$$

everywhere on S^2 . This illustrates that Gauss curvature is intrinsic: different coordinate systems and different expressions for the first fundamental form all encode the same geometric quantity K .